
Fallacies in argumentation on the Web

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1 Introduction

Argumentation is the apparent form of any rational communication. Its quality, soundness, validity is important to convey and receive the right message. As it is the sole process that occurs in any kind of communication, it is pervasive yet not easy to capture for flaws. The flaws that affect the logical entailment in an argument and distort its rationality are called fallacies. It requires a systematic and sound thinking to build good arguments and correctly analyse the existing ones.

As with most of the text analysis tasks in computer science, the automated analysis / detection of arguments is an emerging and challenging topic while it is also a challenge for human speakers.

While argument detection is in a relatively immature state, fallacy detection, which can be regarded a subtask of argument mining, is almost not-touched. Indeed, to our knowledge, there are no computational studies except a few position papers; one that proposes an argument markup language to be useful for fallacy identification [Gibson et al., 2007] and another study that designs for the same purpose a formal language, inspired from automated theorem proving [Reed and Koszowy, 2011].

In this work, we study fallacy detection in texts from user-generated web content. We report our fallacy annotation study with both expert and non-expert annotators, as well as a thorough treatment of the data in the form of our annotation guidelines and a reliability analysis.

2 Fallacy Types

There is not a clear-cut distinction among the types and scopes of fallacy in the field of argumentation theory; the definitions and descriptions are diverse and, in a sense, subjective. Still, there is a form of consensus in many way and there are kinds of fallacies that are widely-accepted, which is natural as the fallacies are the key factors of our daily lives and discussion and the topic is studied for thousands of years in a track of philosophy.

The fallacy types we study and look for in the data to annotate are presented below with definitions from our core bibliography [Sahlane, 2012, Tindale, 2007, Govier, 2010, Damer, 2008] along example texts. There are a total of 20 fallacy types and are grouped for their scope-proximity (such as the context of the flaw (cause or effect) or the direction of attack (emotions, persons..)).

2.1 Ad Hominem Fallacies

This category contains only one fallacy type.

2.1.1 Argumentum ad Hominem

Attributing to or attacking the personal traits, past or intention of the opponent; developing the argument by making use of the motivation, situation or commitment of the opponent.

[Sahlane, 2012]:

p463: quoting (van Eemeren and Grootendorst 1992:212), “doubting the expertise, intelligence, or good faith of the other party”.

p463: quoting (Copi 1986; cited in De Wijze 2003:6), “Ad hominem fallacy, thus, occurs in such cases where the ‘premises are logically irrelevant to, and therefore incapable of establishing the truth of, their conclusions.’”

example

“I have no interest in Professor Lay’s opinions. I’d be surprised if any piece of her sculpture has even “placed” in an art show. Have you ever seen any of her junk?” (Damer08:199)

“Since you aren’t a member of a sorority or fraternity, you’re in no position to tell us how we should or shouldn’t treat our pledges.”(Damer08:201)

Examples from Actual Data

“Grace-Marie Turner president of the Galen Institute, a nonprofit research organization focusing on free-market ideas for health reform - needs to get into the 21st century and realize that "sex shaming" is not an appropriate method of birth control.” (doc#15473651-s1)

“Vegeterians and their Hezbollah-like splinter-faction. the vegans, are a persistent irritant to any chef worth a damn. [on the question as to whether people should become vegetarian or not]” (doc #5315)

“We should all call out faulty science wherever we see it, including the blind belief without any evidence beyond the faulty models that humans are largely responsible for climate change. [on the question as to whether the human activity is primarily responsible from the global climate change or not]” (doc #2904)

“The incredible list of supposed horrors that increasing carbon dioxide will bring the world is pure belief disguised as science. [on the question as to whether the human activity is primarily responsible from the global climate change or not]” (doc #2900)

***caution*:**

Ad hominem can be mixed with straw man or red herring. We present the following annotation as an example:

“Or you could stop being so selfish to the child and send it out into the world so it can learn the importance of society and how to in[t]eract with it. (doc#902-s2)” One annotator finds ad hominem and the other annotator thinks the sentence has straw man due to the word ‘being selfish’. There is a very implicit diversion of the topic so we cannot say it is directly a fallacy of straw man but there is a direct attack on the opponent; the sentence has the fallacy of ad hominem. Thus, we need to distinguish between the attack and the diversion. It depends on the aim of the writer/utterer. It might be the case that both diversion and attack is aimed but usually if there is an attack, there is a slight diversion. We may consider possible replies, for example. Is it likely to move towards a different (diverted) topic or towards self.defense?

Ad hominem can be confused with the fallacy of appeal to emotions. In fact, they are close. We should be careful about the distinction and investigate the writer’s intention and tools. Is the writer appealing to a general audience through some emotions? Is the writer supporting her own claim or refuting the opponent by using / utilising the opponent’s characteristics / properties?

Let’s look at the following sentence: “A Godless discussion in an Atheist Newspaper, so predictable.(doc#15478226-s1)”. One annotator finds appeal to emotion and the other annotator finds ad hominem. There is a direct attack on the opponent; here the writer makes use of the source where the opponent speaks. Yes, there is an appeal to emotion but this comes from the meaning of the words. The intention of attack prevails rather than being emotive.

2.2 Hasty Generalisation Fallacies

Arriving at a conclusion (from general to specific and vice versa) without a comprehensive evidence. [Sahlane, 2012] calls this ‘secundum quid’.

2.2.1 Biased Sample

Developing the argument with the support of subjective sample or illustration which may not cover most of the cases in question.

[Govier, 2010]:

p275: “The [biased] sample is selected in such a way that it is bound either to underemphasize or to overemphasize the characteristic being studied.”

example

“Best-tasting low tar I’ve tried,” report Merit smokers in latest survey. Taste Quest Ends. Latest research provides solid evidence that Merit is a satisfying long-term taste alternative to high tar cigarettes. Proof: The overwhelming majority of Merit smokers polled feel they didn’t sacrifice taste in switching from high tar cigarettes. Proof: 9 out of 10 Merit smokers reported they continue to enjoy smoking, are glad they switched, and report Merit is the best-tasting low tar they’ve ever tried. Merit is the proven alternative to high tar smoking. And you can taste it.” (Govier10:276 quoting from an ad)

“Alternative energy is run by tax payer dollars and cannot sustain itself. Look at Solyndra [a company]. The government [US] gave it 528 million and it went bankrupt. This is why it will never replace fossil fuels. [on the question as to whether the alternative energy can replace the fossil fuel or not]” (doc #cID5585)

Composition/Division

Generalising the properties of the part to the whole or claiming that some property of the whole also holds for the part.

[Govier, 2010]:

p278: “In the fallacy of composition, a conclusion about a whole or group is reached on the basis of premises about its parts or members. The reasoning goes from the smaller unit to the larger one [...]. In the fallacy of division, a conclusion about a part, or member, is reached on the basis of premises about the whole, or group. The reasoning goes from the larger unit to the smaller one.”

example

“Dutch people are tall. Therefore, Holland is a tall country” (Govier10:279)

“The group reached its own decision and was in that sense autonomous. Therefore, every member of the group reached his or her own decision and was in that sense autonomous” (Govier10:280)

Composition/division can be a special case of biased sample. While analysing actual arguments, there is no clear-cut distinction between these two. Therefore we treat composition/division as a biased sample fallacy.

#cID1448 [on becoming vegetarian] “Some of the healthiest people on earth do not eat dairy, and the nations with highest levels of obesity are highly meat-eating nations.”

Composition/division usually involves a generalisation over a single member while biased sample involves multiple members as its object. The previous example was labeled as composition/division; however, it has been found to be a biased sample during the disagreement study/curation. ‘The whole’ is not precise so we couldn’t directly say it has a composition/division fallacy.

2.2.2 False Analogy

Drawing a conclusion using an analogy that actually does not fit in the topic in question.

[Govier, 2010]:

p340: “Such[such loose and far-fetched analogies that it is impossible even to classify them as a priori or inductive] loose uses of analogy are often discussed as instances of the fallacy of faulty analogy. They involve an appeal to similarities that are highly superficial and give no real support to the conclusion sought.”

[Damer, 2008]:

p151: “Assuming that because two things are alike in one or more respects, they necessarily are alike in some other important respect, while failing to recognize the insignificance of their similarities and/or the significance of their dissimilarities.”

example

“If one were to listen to only one kind of music or eat only one kind of food, it would soon become boring or tasteless. Variety makes eating and listening exciting and enriching experiences. So it seems to me that an exclusive sexual relationship with only one partner for the rest of one’s life—that is, marriage—does not hold out much hope for very much excitement or enrichment.” (Damer08:152)

“If there is true peace, the Jewish right to continue to reside in the West Bank should not be abrogated, just as Israel’s 1.3 million Arab citizens mainly of Palestinian descent will have the right to remain in Israel even if a Palestinian state is created. [on the question as to whether the Israeli settlements in Palestinian territories is an obstacle to peace process or not]” (doc #4120)

2.3 Causal Fallacies

Developing arguments that claim false causes, confuse cause and effect, mistake correlation for cause, assume the preceding event is the cause for the successor, assume the claim true due to not knowing the exact reason or rephrase the claim as if it is the cause.

2.3.1 False Cause

Developing arguments that claim false causes, mistake correlation for cause, assume the preceding event is the cause for the successor or confuse cause and effect.

[Govier, 2010]:

p308: “The fallacy of objectionable cause occurs when someone argues for a causal interpretation on the basis of limited evidence and makes no attempt to rule out alternative explanations of the event.”

[Damer, 2008]:

p180: “[Post hoc fallacy assumes] that a particular event, B, is caused by another event, A, simply because B follows A in time.”

[Govier, 2010]:

p305: “[Confusing correlation and cause] is a mistake to infer from a premise that X is positively correlated with Y, a conclusion to the effect that X is the cause of Y. This correlation is compatible with Y causing X, with the relationship being a coincidence, and with the existence of a third factor that is the underlying cause of both X and Y.”

example

This example is said to be hosting post hoc fallacy but it also contains correlation mention underneath: “If it were discovered that most elementary schoolteachers have children of their own, it might be concluded either that teaching stimulates an interest in parenthood or that being a parent stimulates interest in working with children.” (Damer08:184)

This is a better example for post hoc: “Ever since we quit going to church, business has been getting worse. If we want to keep from going completely bankrupt, we’d better go back to church.” (Damer08:181)

This one is given for illustrating false cause (misidentifying the cause (Tindale07) / objectionable cause (Govier10)) fallacy. However, there are attributes to both correlation and consequences: “The decision of large numbers of women in the last decades of the twentieth century to abandon the home and instead pursue careers in the workplace has led to a decrease in the birthrate, since fewer are making the decision to stay home and be mothers.” (Tindale07:p180)

This is given as an example for confusion of cause and effect: “One prison inmate says to another: ‘Governor Kaine always seems to know when we’re having a good meal. He times his annual inspection visit here on the one day of the year that we have steak.’” (Damer08:182)

2.3.2 Begging the Question

Developing arguments that rephrase the claim as if it is the cause.

[Tindale, 2007]:

p73: “Someone begs the question when she assumes in the premises the very thing she is asserting in the conclusion. Since the same statement cannot be used to prove itself, the argument is circular.”

example

“A heavier-than-air craft could never fly because in order to lift up and travel over distance a machine would have to be lighter than the environs surrounding it.” (Tindale07:74)

2.3.3 Appeal to Ignorance

Also *Argumentum ad Ignorantiam*.

Developing arguments that assume the claim true due to not knowing the exact reason.

[Sahlane, 2012]:

p472: quoting (van Eemeren and Grootendorst92:212-216), “A discussant commits an argumentum ad ignorantiam fallacy ‘by concluding that a standpoint is true just because the opposite has not been successfully defended’.”

example

“Angels exist, because no one has ever proven that they don’t.” (Govier10:175)

“Predictions of future climate come from computer models, which are very incomplete approximations of the behavior of the real climate system... The predictions of future climatic changes are hypotheses, not scientific facts... Whatever the threat of climate change to humanity, it is most likely to be natural - not man-made. [on the question as to whether the human activity is primarily responsible from the global climate change or not]” (doc #2907)

[DO: DO: *False cause, post hoc, correlation-cause confusion and cause-effect confusion should be merged under the name false cause or another suitable one. They are interrelated and one type of fallacy might contain at least another one in this class. In most of the cases, correlations, effects and causes are mentioned altogether.[We did this indeed (DO, 28.07)]*

We can consider adding ‘argument by innuendo’ where the claim is presented via implicatures or some evidence clues (Damer08:129) yet it is hard to detect automatically and even manually.]

2.4 Appealing to Consequences Fallacies

Also *argumentum ad consequential*.

Arriving at a conclusion by attributing to the effects or consequences in a way that skips the reasons and that might appeal to emotions (threat, joy, fear..) as well.

2.4.1 Slippery Slope

Developing the argument by appealing to a chain of consequences which generally end up with a negative outcome.

[Damer, 2008]:

p1185: “Assuming, without appropriate evidence, that a particular action or event is just one, usually the first, in a series of steps that will lead inevitably to a specific, usually undesirable, consequence.”

example

“Unbridled passion following the wake of birth control will create a useless and effeminate society, or worse, result in the complete extinction of the human race.” (Govier10:310)

caution:

We should be careful about the natural / expected consequences and the exaggerated consequences of the event in question. Those that are natural do not generally lead to a fallacy of appealing to consequences. However, if the arguer is exaggerating them or skipping the expected results or positive sides, then there is a type of manipulation and this is mostly a fallacy. We give the following piece as an example, where the writer / speaker thinks that social networking sites are not good for the [their] society, saying “In the internet era, it appears there are more people interested in spewing hate than in countering it. On the social networking sites and on YouTube, inflammatory, hate-filled content overwhelms the limited efforts to promote tolerance and to teach diversity. And, as we have seen, hate speech inspires violence (doc #4330)”. One annotator labelled the last sentence to be containing the fallacy of appeal to consequences. However, what the writer say is true, if there is hate speech, it invites violence, which is itself a type of violence indeed. This is a factual information and the writer delivers her opinion about the progress of those social networking sites. It is not necessarily flawed to present current state /

factual expectations. This requires real-world knowledge.

We see the same mistake in the following sentence that says “The big question is whether this will be forced on us by the inexorable economic chaos from depletion of our primary non-renewable resources, oil, natural gas and coal, or by our adapting to this inevitability in a thoughtful and organized manner. [on the question as to whether the alternative energy can effectively replace fossil fuels] (doc #7132)”. It is widely-known, at least historically, that energy deficit brings economic chaos so what the writer is saying may not be regarded as some dissuasive and exaggerated list of consequences. And yet another such case where one annotator thought there was an appeal to bad consequences while the consequences mentioned are widely known: “Our actions are effecting our world and animals and if nothing is done now while we still have time... Our world will never be the same! [on the effect of human activity over global climate changing](doc #cID3140)”. However, in this sentence, there is linguistic complication and wording has a slight appeal to emotions, maybe this is what brings the confusion.

2.4.2 Sunk Costs (Argument from Waste)

Assuming the possibly wrong/false claim/action true since the reverse is expensive.

[Sahlane, 2012]:

p478: citing (Walton02:475), “The ‘sunk costs’ fallacy or ‘argument from waste’ is known as ‘the Concorde fallacy’ (the British and French governments remained committed to the supersonic jet project, even after it became clear that it would fail, simply because so much money had already been spent on it by that point).” and he adds that “[t]he fallaciousness of ‘sunk costs’ arguments, however, remains context-dependent.”

example

See the definition by [Sahlane, 2012] above (Section 2.4.1).

2.4.3 Appeal to Force

Also *argument ad baculum*.

Justifying the claim by using the consequences of the reverse as a threat as if it would make the reverse false.

[Sahlane, 2012]:

p469: quoting (Bunnin and Yu 04:48), “Argumentum ad baculum involves ‘an attempt to win assent for a conclusion by appealing to force or by issuing threats concerning the consequence that will follow if the conclusion is not accepted’.”

example

“You had better believe in God, because if you do not, when you die you will go straight to Hell and suffer eternal damnation.” (Govier10:170)

[DO: *Slippery slope is a type of appeal to consequences however long the chain of succeeding events are. We can omit it and consider only consequences. It is also fairly similar to appeal to force/threat. Interestingly, both Govier and Damer consider slippery slope under causal fallacies.*]

2.5 Pleading Fallacies

Developing arguments that skips reasoning and put emphasis on emotions including pity, trust, confidence, belief. The justification of the claim is done via the promised gain from the emotions like confidence in common sense or authority or pity, guilt etc even though these emotions conflict with the truth.

2.5.1 Appeal to Popularity

Also *argumentum ad populum*.

Justifying the claim by another claim that many people or common sense accept it. Called 'appeal to common opinion' by [Damer, 2008] and said by [Govier, 2010] to be also known as 'bandwagon appeal' or the fallacy of 'jumping on a bandwagon' or 'ad populum'.

[Govier, 2010]:

p380: "Urging the acceptance of a position simply on the grounds that a large number of people accept it or urging the rejection of a position on the grounds that very few people accept it."

example

This can also be an example of 'lack of evidence' but is given for illustrating appeal to popularity in (Damer08:105), "Marijuana can't be all wrong. According to a recent Gallup survey published in yesterday's Wall Street Journal, more than 60 percent of the adult American population sees nothing wrong with it."

2.5.2 Appeal to Emotion

Appealing to rather strong emotions that affect the audience or the opponent upon making a decision about the claim. It has rather a convincing or manipulating tone and the reasons and the evidence usually stay covered behind the emotions.

[Damer, 2008]:

p111: "Attempting to persuade others to accept a position by exploiting their emotions instead of presenting evidence for the position." [Damer, 2008] mentions the related emotions such as pity, flattery, praise, guilt, loyalty, shame while [Govier, 2010] adds envy, joy, fear, desire, compassion, empathy, sympathy, hatred, insecurity, pride, excitement.

example

"You had better believe in God, because if you do so, when you die you will go to Heaven and experience eternal bliss." (Damer)

"Remember, keeping your kids away from knowledge that you don't like is a moral crime." (doc#283-s1)

"This dramatic and unprecedented Israeli demonstration of good-will was rejected by the Palestinians outright, even before it was officially announced... Building inside existing communities - including construction of schools, synagogues and medical clinics that allow their residents to lead a normal life - does nothing to change the facts on the ground nor does it prejudice whatever might be agreed upon in a final peace agreement. [on the question as to whether the Israeli settlements in Palestinian territories is an obstacle to peace process or not]" (doc #4122)

2.5.3 Appeal to False Authority

Developing the argument by leaning on the false expert opinion or unknown authority as if they compensate for the lack of concrete evidence. There is an appeal to loyalty or confidence feelings as well. We consider the appeals made to tradition also under this category, where an action is claimed to be pursued just because it is traditionally done so while omitting the other and current effects of it.

[Damer, 2008]:

p102: “Attempting to support a claim by appealing to the judgment of one who is not an authority in the field, the judgment of an unidentified authority, or the judgment of an authority who is likely to be biased.”

example

“Everything the president says is true. The president says that Iraq has weapons of mass destruction. Therefore, Iraq has weapons of mass destruction.” (Tindale07:151)

This is an example (rather contrapositive) for ‘appeal to tradition’ that we encountered during our annotations: “There should not be one state because they have had separatism for far too long. [on the question as to whether one state solution is an acceptable one to the Israeli-Palestinian conflict]” (doc #cID5808)

caution:

Not every unquoted resource should be labelled ‘false authority’. In daily or informal speech, people can quote the research, scientific findings or newspaper pieces without proper citation. However, in an argumentative discourse if one party is making use of some unknown / unspecified research as the main refutation tool of her attack rather than a supporting fact, then it is more likely to be fallacious. For example, the following sentence that supports not being vegetarian “Research shows beef’s high-quality protein and essential nutrients make good diets better, improving overall nutrient intake and diet quality, and contributing to positive health outcomes like weight management. (doc #5314)” is annotated to be having the fallacy of ‘appeal to false authority’. However, this is only some side information, not the main attack point. We cannot say it is directly appealing to some unspecified resource and indeed it is from some widely known facts. We should distinguish that extra, precious information and the widely-known knowledge.

2.6 Diversion Fallacies

Developing the argument by drawing the attention to a different topic or a neglectable aspect (somewhat irrelevant and usually deliberately), provoking or underrating and despising the opponent viewpoint.

2.6.1 Straw Man

Misinterpreting or disinterpreting the opponent claim and developing the counter argument based on this diversion.

[Govier, 2010]:

p176: “A fallacy committed when a person misrepresents an argument, theory, or claim, and then, on the basis of that misrepresentation, claims to have refuted the position the person has misrepresented.”

[Tindale, 2007]:

p20: “The Straw Man fallacy involves the attribution or assumption of a position, which is then attacked or dismissed. The problem is that the position dismissed by the argument is not the real ‘man’ or ‘person’, but a caricature of the real position held.”

example

“The proposal of my distinguished colleague[politician] from the state of Virginia wants to cut our military budget in a way that fails to support our troops in the Middle East and would undercut our defense posture around the world. I say that we must not do something that would make us into a second-rate military power and thus prevent us from keeping our commitments abroad.” (Damer08:204)

“Breakthrough technologies, such as cellulosic ethanol, are theoretically attractive - but don’t exist. [on the question as to whether alternative energy can effectively replace fossil fuels or not]” (doc #7138)

***caution*:**

Straw man can be mixed with ad hominem. We present the following annotation as an example:

“Or you could stop being so selfish to the child and send it out into the world so it can learn the importance of society and how to in[t]eract with it. (doc#902-s2)”

One annotator finds ad hominem and the other annotator thinks the sentence has straw man due to the word ‘being selfish’. There is a very implicit diversion of the topic so we cannot say it is directly a fallacy of straw man but there is a direct attack on the opponent; the sentence has the fallacy of ad hominem. Thus, we need to distinguish between the attack and the diversion. It depends on the aim of the writer/utterer. It might be the case that both diversion and attack is aimed but usually if there is an attack, there is a slight diversion. We may consider possible replies, for example. Is it likely to move towards a different (diverted) topic or towards self.defense?

2.6.2 Red Herring

Asserting provoking and irrelevant claims; pointing to a different subject.

[Govier, 2010]:

p176: “A premise or remark that is irrelevant to the conclusion or issue being discussed, so that it tends to distract people and lead them away from the topic at issue.”

example

The example given by Tindale08 to illustrate red herring, while it also contains appeal to emotions: “Many of your readers seem devastated by the callousness of the annual seal hunt and the culling of seal pups, but their concern is misapplied when they so blithely overlook the real crime being perpetrated every year in this country in the slaughter of thousands of the unborn. Abortion accounts for more deaths of innocents than many wars, but still society does nothing.” (Tindale07:p49)

***caution*:**

It is important to note the difference between straw man and red herring fallacies. The former contains mis-interpretation of the opponent claim while the latter only draws attention to something different. To illustrate, suppose the opponent claims that public education is better than private education. A person-A attacks to this saying “No, it is not good. Socialism has no good for our children.”. Here person-A presents the opponent claim in an over-interpretation; thus, this is straw man fallacy. Another person-B argues, saying that “No, it is no better. We need to have a better government to achieve this and the current one has no intention to improve education”.

Here person-B claims the opposite but adds new information to the discussion, diverts the topic of education to government.

2.6.3 Appeal to Sarcasm and Ridiculing

Developing the argument in a sarcastic / cynical / ridiculing way against the opponent view or to underrate the issue via some humour.

[Damer, 2008]:

p210: “Resort to humour or ridicule is [to] inject humor or ridicule into an argument in an effort to cover up an inability or unwillingness to respond appropriately to an opponent’s criticism or counterargument.”

example

“Well, if I were elected, about half the members of Congress would drop dead of heart attacks, and half of my problem would be solved from the outset.” (Damer08:210)

“That is how [by having a healthy diet that includes animal food] one recovers from vegetarianism and builds lasting health and strength.” [on question as to whether people should be vegetarian or not] (doc #5313)

“There is no escape from dependence on slaughtered animals , not even for really good vegan folks who feel wonderful about themselves as they finish their vegan meal...” [on question as to whether people should be vegetarian or not] (doc #5310)

“I mean honestly, you think your cars run off of rainbows and glitter?” [on question as to whether people should be vegetarian or not] (doc #CID3205)

***caution*:**

This fallacy can be easy to capture if the required background knowledge is not special. For example:

“If you want to chat with your invisible, supernatural friend [on practising religion] do so off school property and on your own time. (doc#1546-s2)”

Both annotators think there is a fallacy of appeal to sarcasm. The annotators agree and are right. The writer presents his reason for objecting in a humiliating way. However, this is not to be confused with ad hominem. Here nobody is blamed, no claim/reason is interpreted / re-dressed using the opponent’s actions or personality. The only thing is that there is a style that degrades the opponent claim and promotes her own claim but at the same time the claim itself is presented / uttered as well (in this case, the writer claims that in the schools there shouldn’t be any religious practice because it involves some supernatural things and this chatter should be done in some personally space and time. We see, the claim is there and is supported by some degrading definitions. And no new claim is presented, the writer is on the topic. Yes there is a slight diversion about the existence of religion but it can just be omitted as the writer’s own opinion and the discussion can go on on the topic of praying in the schools.).

3 Preliminary Annotation Study

We have a collection of around 341 texts which were previously collected and annotated for argument components by HabernalGurevych15-ref[Habernal and Gurevych, 2012]. The texts are user comments posted to forum discussions or articles in the education domain. They are usually written for the purpose of debating, thus are rich in argumentative discourse and yet they have an informal style, which possibly may lead to many fallacies and relative difficulty in detecting them. [Habernal and Gurevych, 2012] presents an argumentation model ([Toulmin, 2003]) and annotates the arguments in these texts according to that model, where the components are premise, claim, backing and rebuttal. Thus,

the texts are already labelled for their arguments (each text has one argument) while there can be other arguments related to side/off-topic issues or even sub-arguments.

There are two annotators, namely A1 and A2. For this preliminary study, we analysed the texts sentence by sentence so the unit of annotation is sentence as opposed to our intention to find the fallacies only in the labelled arguments in the text. The categories are the fallacy types introduced above, which is a total of 20 types. [Much high for a plausible annotation task]. It is allowed to label a sentence with more than one labels. The annotators used a shared spreadsheet to record their answer, none seeing each other's answer until the discussion.

Inter-annotator agreement is calculated using Krippendorff's alpha measure with MASI distance metric developed by Passonneau06-ref[Passonneau, 2006]. We used dkpro-statistics library (Meyer-et al14-ref[Meyer et al., 2014]) that implements all of these metrics. This distance metric, said to be derived from Jaccard's coefficient, treats annotations (both single and multi-labels) as sets so each member label is an element of the set and thus each member label contributes to the overall agreement computation. In addition, we also tried its previous version 'distance metrics for sets' by Passonneau04-ref[Passonneau, 2004] as well as the nominal distance metric.

3.1 commentset1

Out of 341 texts, we selected 15 texts which are comments to articles by users. There was a total of 86 sentences. Two annotators annotated each sentence of each document for fallacies using the guideline presented in the previous fallacy types chapter.

The fallacy types as well as their encodings are shown in the table below.

labelid	label name / fallacy type	labelid	label name / fallacy type
1	argumentum ad hominem	11	slippery slope
2	hasty generalisation / specification	12	sunk costs
3	biased sample	13	appeal to force
4	anecdotal arguments	14	appeal to popularity
5	composition / division	15	appeal to emotion
6	false analogy	16	appeal to false authority
7	false cause	17	straw man
8	begging the question	18	red herring
9	argumentum ad ignorantiam	19	appeal to sarcasm / ridiculising
10	argumentum ad consequentiam	20	trivialising / trivial objections

Table 1: Fallacy types (category labels) used in annotating commentset1

It took on average about 2 hours to complete annotation of this 15 documents.

3.1.1 Agreement results

Below we present our agreement results.

distance metric	agreement value
MASI distance metric	-0.0233
set distance metric	-0.0234
nominal distance metric	-0.0309

Table 2: Agreement results based on Krippendorff's alpha of the annotations in commentset1

The results are apparently low so the annotations are surely not reliable. The problems are discussed verbally. We briefly explain the problematic factors here as:

- The texts are informal and this informal tone is sometimes confused for logical flaws or linguistic ambiguities.
- One of the annotators interprets some of the fallacies different, somewhat different than their average definition, which introduces errors. Additionally, one of the annotators handles the text in a very detailed, picky way, which causes an intoleration to the usual, very implicit flaws in daily speech.
- The number of categories, which is 20, is reasonably high to choose among and remember.

3.1.2 Label-based analysis of the annotations

We conduct this study to understand the capturability of our fallacy types. This will also help us distill our guidelines / set of types better.

Overall frequency of the labels

Below we present the number of times any of the annotators used the corresponding label (or set of labels) for a sentence. As we have 86 sentences and two annotators, there is a total of 172 annotations.

labelid	label name / fallacy type	# of times used
1	argumentum ad hominem	6
2	hasty generalisation/specification	1
7	false cause	1
10	appeal to consequences	4
11	slippery slope	1
15	appeal to emotion	5
16	appeal to false authority	2
17	straw man	7
18	red herring	4
19	appeal to sarcasm / ridiculising	4
na	no fallacies	116
1, 17	ad hominem + false cause	1
1, 18	ad hominem + red herring	1
13, 18	appeal to force + red herring	1
14, 7	appeal to popularity + false cause	1
15, 7, 18	appeal to emotion + false cause + red herring	1
18, 9	red herring + argumentum ad ignorantiam	1
2, 16	hasty generalisation + appeal to false authority	3
2, 16, 7	hasty generalisation + appeal to false authority + false cause	2
2, 17	hasty generalisation + straw man	1
6, 7	false analogy + false cause	2
7, 18	false cause + red herring	1
7, 18, 19	false cause + red herring + appeal to sarcasm	1

Table 3: Frequency of the labels in the annotations in commentset1

In the Table 3, we see:

- no-fallacies is the most used label. then, straw man, ad hominem and appeal to emotion are the most used fallacy types.
- red herring, false cause, false authority, hasty generalisation frequently occur together, while they are not much detected alone.
- appeal to emotion is pervasive.
- appeal to emotion, straw man and red herring is the most used triple.

The fallacy types that are not used in these annotations are:

- biased sample
- anecdotal arguments
- composition / division
- begging the question
- sunk costs
- trivialising (trivial objections)

Agreement rank of the labels

Below in Figure 1 we present the contingency matrix.

	15, 17, 18	11	10	na	2, 7, 18	15	15, 7, 19	18	2, 17	13, 17	1, 17	15, 10	2, 16	7, 18, 19	2, 16, 7	18, 9	1, 14, 7	6, 7	1, 18	Σ	
16	2																			4	
15, 17, 18	2					2														6	
11					2															2	
10			2																	8	
na		2		2	148	2	2	8	6	2	6	2	14		2	4	6	2	2	232	
2				2	2	2											12	2	4	2	
7, 18					2															2	
15			2			8														10	
19																				8	
15, 7, 18											2									2	
18																2				8	
2, 17																				2	
17																				14	
13, 18																				2	
1, 17																				2	
15, 10																				4	
2, 16																				6	
7, 18, 19																				2	
2, 16, 7		2																		4	
18, 9									2											2	
1																				12	
14, 7																				2	
6, 7																				4	
7																				2	
1, 18																				2	
Σ	4	6	2	8	232	2	2	10	8	2	8	2	14	2	2	4	6	2	4	2	344

Figure 1: Contingency matrix.

From the matrix we see that:

- **no-fallacies**: 37 sentences were agreed on being non-fallacious; the most used and confused label. It is confused / co-occurred mostly with ad hominem and straw man.
- **appeal to consequences**: 1 sentence was agreed by both annotators to have the fallacy of this type (doc#1414-s7: “Ultimately, we as a society lose.” : apparent due to the adverb ultimately). It was co-occurred once with slippery slope, which is a variant of appeal to consequences indeed (doc#45-s1: “It can ruin your child’s whole life.”) and once with no-fallacies (doc#4068-s1: “I have been advised to put my money in private schools, or else save it for attorneys later.”). This label is used together with appeal-to-emotion by one of the annotators in two sentences.

3.1.3 Disagreement analysis of the annotations in commentset1

We present the analysis of disagreements on our annotations of commentset1, whose agreement values we presented in the previous section.

doc#45

S0Depriving your child of a basic education is a form of child abuse. S1It can ruin your child’s whole life.
claim: S2Home schooling should be illegal unless rebuttal: the parent can demonstrate that they are providing the same level of education as a public school. S3There should be a core national curriculum and testing to ensure children are achieving at least a basic level of education.
premise: S4In an increasingly complex, global technological society, all people need to have a basic understanding of science, technology and local and global culture, just to be able to function and make informed decisions.
claim: S5I don’t see any need for home schooling any child unless rebuttal: the child has special needs or learning difficulties. S6If public schools are under-performing, then the public education system needs to be improved.
premise: S7Public education in the US seems to be a self-perpetuating disaster, with ignorant, uneducated, unqualified people on school boards deciding what children should learn.

A1 finds two fallacies, agreed by A2. And A2 finds at least 5 more.

(s0): Both A1 and A2 find appeal to emotion. A2 also finds red herring due to 'child abuse' and straw man due to 'deprivation of education'.

In the context, the topic 'child abuse' is not red herring but the misinterpretation of homeschooling as 'deprivation of child from education' is fallacious, it has straw man fallacy.

(s1): A2 finds appeal to consequences due to 'ruin the whole life'.

It is exaggerating and appeals to extreme consequences.

(s5): A2 finds hasty generalisation due to 'no need for homeschooling'.

This might be a generalisation but it is the writer's opinion and we cannot say it is a fallacy in that regard.

(s7): A2 finds false cause due to 'bad education because of unqualified deciders', as well as red herring due to shifting the topic to the quality of education.

The cause of 'disastrous' education can have a number branches while the writer presents only one as if it was the only cause; so this is fallacious, fallacy of false cause. The topic is shifted to the quality of education but it does not continue arguing any more so the sentence does not necessarily have red herring.

doc#283

S0Well this has been fun but I have to go back to work.

premise: S1remember, keeping your kids away from knowledge that you don't like is a moral crime.

S2Have a nice day.

Both annotators agree that (s1) has appeal to emotion and A2 finds two more fallacies in it.

(s1): The two annotators find appeal to emotion due to 'moral crime'. Additionally, A2 finds straw man due to 'keeping children away from school' and red herring due to 'moral crime'.

The writer appeal to the emotions of guilt by labelling the action in question as 'moral crime'. It is not necessarily red herring, not a completely new subject is introduced. Misinterpreting homeschooling as 'keeping away from knowledge' is apparently diversion and it is the fallacy of straw man.

doc#718

S0Interesting. S1I commend your commitment and I wish you all the best for your sons. rebuttal: S2I suspect you will probably do a better job of educating them than most schools would.

premise: S3I can't help wondering, though, what the fallout is likely to be in terms of their ability to socialise with other kids. S4Whatever home schooling does for you intellectually, I can't think it can have that great an influence on their ability to fit in with other kids.

S5Prepared to be proved wrong though....

A1 finds no fallacies and A2 finds sarcasm in the first three sentences.

(s0, s1, s2): A2 finds ridiculising / appeal to sarcasm due to the words 'wish you best', 'suspect you do better'.

There is a linguistic and contextual ambiguity. The writer gives a reply to some previous post and it is not sarcastic, it is most probably a part of conversation. A2 must have missed this point and over-interpreted.

doc#927

premise: S0Ours will be that the State Education system is totally ruined, due to many years of being systematically abused by left-wingers.

claim: S1I wouldn't trust a state school to train a dog, much less raise a child.

A1 senses some sarcasm, indicating that it is not necessarily a fallacy and A2 finds several fallacies.

(s0): A2 finds false cause, appeal to emotion and red herring due to 'education ruined by left-wingers'.
As in the previous case, ruin of education can have several causes. Although this requires historical knowledge, one shouldn't represent one party as the only reason. There is a fallacy of false cause. But we cannot say the sentence appeals to emotions or diverts the topic, it is still on the topic of education.

(s1): A2 finds red herring due to 'train a dog'.
It is not diversion; it is rather appeal to sarcasm.

doc#1220

claim: S0Nobody needs to be practicing religion at a public school. premise: S1The whole point is to separate education from religion. premise: S2If you want your kid learning and praying in the same place then send them to a religious school.

A1 finds no fallacies and A2 finds two fallacies overall.

(s0): A2 finds hasty generalisation due to 'nobody needs' and straw man due to 'practising religion'.
The writers conveys her opinion how general it is; we cannot say it is an inference so there is no fallacy. Also practising religion is not misinterpretation; prayer in schools indeed is a practice of religion.

(s1): A2 finds straw man due to comparing education and religion in the first place.
The writer does not regard the two in the same class or substitute one for another; she just says they should be separated. This is not a misinterpretation of the topic at hand.

doc#1403

claim: S0No prayers in schools! S1Offend everybody equally!!

premise: S2They do not belong. S3Pray on your own time.

A1 finds appeal to sarcasm and A2 finds several different fallacies.

(s0): A2 finds red herring due to the exclamation at the end of the sentences.
Not all exclamations are provocation; this can be a clue how the writer empowers her claim/belief, not a provocation. There is no fallacy in this sentence.

(s1): A1 finds appeal to sarcasm due to 'offending everybody' and A2 finds appeal to emotions and red herring for the same clause. Additionally, A2 finds appeal to force for the writer says the opponent action will be offensive if taken.
The verb offend appeals to emotions rather than being sarcastic or provocative / diversive. A2 is right, the writer is appealing to some dissuading/threatening result.

(s3): A2 finds straw man and ad hominem fallacies.
This is a suggestion, not necessarily an attack or misinterpretation since the topic is already praying.

doc#1414

S0There are many intricate details when making decisions in regards to drawing the line when separation is required. S1That said, should the decision have been made by one person I would understand and cut that person some slack. S2None of us are perfect. S3What bothers me here is that this appears to have been a group decision. S4That is scary for rest of us. claim: S5Please stop, reverse your decision and realize that religion, all religions, have no use in our education system, our government and technically speaking (even those whom are dedicated to their faith) 98% of the hours with a calendar year. premise: S6Our society just isn't gear for that. S7Ultimately, we as a society lose.

Both annotators agree on appeal to consequences in the last sentence and A2 finds some more fallacies in the preceding part.

(s4): A2 finds appeal to emotion and appeal to consequences due to writer's mention of some scary condition. *This is not necessarily an appeal to consequences. Indeed, he reports his opinion about some situation, not directly related to the argument or not presenting hidden claims per se.*

(s5): A2 finds straw man due to 'religion and education comparison'. *As encountered previously, the writer is on the topic, there is no diversion or misinterpretation.*

(s7): Both annotators find appeal to consequences due to 'ultimately we lose'. *Both are right, the sentence is about the consequences of the opponent claim.*

doc#1467

S0The school board's decision is having the exact opposite effect of what they intended..premise: S1This political correctness is making people less tolerant, creating hard feelings and driving an even bigger wedge between people..premise: S2Thousands of different religions in this world but everyone seems to think that their religion is the only one that counts... backing: S3Religion is basically a gang mentality where people feel they need to belong to a group...claim: S4Get it out of public schools

A1 finds no fallacies and A2 finds several overall.

(s1): A2 finds hasty generalisation due to the writer's comments about the situation of some people. *It is an opinion and indeed not directly related to the argument. There is no fallacy.*

(s3): A2 finds red herring and ridiculising / appeal to sarcasm due to defining religion as a gang and also finds false cause due to the given reason about people commitment to religion. *It is not red herring, there is no diversion but it contains sarcasm and is connected to the claim. So there is a fallacy of ridiculising / appeal to sarcasm. However, the writer is not over or underestimating the reason why people are religious, he is right to some extent so there is not a direct false cause fallacy.*

(s4): A2 finds straw man. *There is no fallacy, nor a misinterpretation. Religion is in the schools with prayers, the writer gives her own comment about it.*

doc#1621

claim: S0As a student, i think schools should definately support prayer in schools . S1No amount of teachers and staff can keep students from throwing up a prayer before class or a test or before they eat lunch. S2As far as a moment of silence during homeroom, I say why not?. backing: S3Children who don't want to pray don't have to, and those who do can have their concentration to do so. premise: S4A moment of silence doesnt require students or teachers to pray; Prayer isnt even mentioned in the words, so why make a big deal of it?. rebuttal: S5I fully understand nonbelievers or nonreligious people who dont pray, but refutation: their moment of silence can be used for relaxation or even sleep if they want.

A1 finds no fallacies and A2 finds two fallacies.

(s4): A2 finds appeal to emotion due to 'a moment of silence, why not'.

The writer is omitting the other details and showing the praying action as a small, harmless thing, appealing to pity.

(s5): A2 finds straw man due to the writer's suggestions to non-believers about what to do in the praying time.

This is not a misinterpretation, this is writer's opinion and has no effect in the claim or attack.

doc#1909

S0Isherwood

premise: Yes, that's true, but the purpose is to give students time to collect thoughts, pray, do homework for the class at hand, etc. backing: Sorry, I can't quote the source, but I read some statistics the other day that students who make time each day for some silence (not necessarily religiously related silence and/or prayer, just silence) do much better in school because people don't realize how much noise dominates our days and inhibits our thought. premise: S1I don't think 2 minutes of silence could possibly hinder our absolutely terrible public education any further than it's current decrepit state.

S2It's all good. S3Don't worry, I agree. S4Trust me, I know. S5First hand.

S6Learn what? S7The BS they half-arsedly try to teach us? S8I attended private school up until age 14, and I have yet to learn anything from our public ed. system.

S9True, but the scenario at the top was obviously over-the-top wrong. S10I just wanted to present a much more common iteration.

Both annotators agree on two sentences though A2 finds some more fallacies.

(s0): A1 finds appeal to false authority due to not quoting but mentioning the research. A2 labels the same problem as lack of evidence, hasty generalisation and biased sampling. A2 also finds false cause / correlation due to 'doing better after some silence since people are not aware of noise'.

The problem about unquoted research relates to false authority rather than hasty generalisation. We cannot say the given reason about silence-success correlation is false because it is a result of the research, not presented by the writer and we already labelled the unquoted research as false authority so the superset of this clause is already wrong.

(s1): Both A1 and A2 finds red herring due to pointing to bad education as a compromise, it shifts the topic and A2 additionally finds ad ignorantiam due to the writer's being not very certain about her claim.

In addition to red herring, this sentence has actually the fallacy of sunk costs / argument from waste. There is ad ignorantiam, though very implicitly and not directly on the topic.

(s2, s3, s4, s5): A2 finds ad hominem and appeal to emotion due to the writer's call to be trusted.

There is a linguistic and contextual ambiguity; this is an informal text and the writer's call is not serious. We can discard these sentences, they are irrelevant.

doc#2203

S0I think the individual schools themselves and the individual needs of your daughters are more important than generalisations about education and single-sex or co-ed schools.

claim: S1However, if you want an opinion from a general education perspective, and you can read up on this, the general rule is that single sex schools are better for the academic performance of your daughters. backing: S2Research is quite clear about this. premise: S3For a variety of reasons, boys in a classroom has an overall negative impact on girls' learning.

A1 finds no fallacies and A2 finds some pleading.

(s1): A1 finds appeal to popularity and false cause due to 'general rule'.

The claim needs to be supported by presenting objective conditions but the writer only says it is general rule, which is unknown. There is no cause presented and it is not necessarily a fallacy of false cause.

(s2, s3): A2 finds hasty generalisation / false cause due to 'various reasons' and lack of evidence due to 'some research'.

The writer presents her own opinion so there is not a fallacy of false cause or hasty generalisation directly. But mentioning some research for support signals appeal to false authority.

doc#2267

backing: S0I am a scientist and I believe in data over anecdote, yet I experienced the beneficial effects of single-sex education myself. S1After co-ed public high school and a freshman year at a co-ed university, I transferred to a women's college and the difference was astonishing. premise: S2Women spoke. S3Assertively. S4Being intellectual and smart was cool. S5Speaking up was cool. S6When I went to law school, and later to grad school, I could always pick out the women who attended women's colleges. S7We all could.

S8Sadly, my college chose to go co-ed years ago. S9It left a hole in my heart. claim: S10Had I a daughter, I would encourage her to find single-sex schools.

A1 finds appeal to emotion once and A2 finds some other fallacies.

(s0): A2 finds ad hominem due to the writer's mentioning of her occupation as a scientist.

This is not a fallacy; indeed, the problem rather stems from A2's different interpretation of ad hominem. Ad hominem contains the attacks or attributes directed towards the opponent person not any person in the discussion. Thus, there is no fallacy in this sentence, though we may say there is a slight taste of appeal to authority because of the attempt to increase credibility.

(s2-s10): A2 finds false cause due to writer's presenting talkative women as a cause of good education. A2 also finds anecdotal fallacy.

This is a personal story, not necessarily stating a factual judgement; the writer only presents her opinion. There is no fallacy in this regard.

(s8, s9): A1 finds appeal to emotion due to 'sadly' and 'hole in the heart'.

These words are not directly related to claim, they are a part of the story.

doc#3158

backing: S0Having sent my children to both public and private schools, claim: I prefer private schools. premise: S1They get a better education and they are safer from acts of violence. S2Public schools have become more interested in political correctness than in education and there is too little control over violence.

Both annotators find no fallacies though A2 senses hasty generalisation, stating to be unsure.

doc#3341

backing: S0I am in no way "Wealthy" however I chose to send my children to private school that has a religion based background. premise: S1Our schools are taking God out of the schools and that is my main deciding factor. S2I am involved as a parent, why is it that we have religion in Jails but not in the public school system. S3And I think its every parents right to chose to educate their child as they see fit. premise: S4The curriculum is much more hands on and students learn more. S5They could do this at public schools but do not. S6So I think your argument needs a lot more supporting facts. S7Seems to me that you would send your child to private school if you felt you had the means. S8Where there is a will there is a way. S9There are grants that can be available too. S10If the government actually spent money on the school system instead of other things that would be different.

A1 finds no fallacies while A2 finds several.

(s0): A2 finds red herring due to the writer's mentioning of her wealth state.

This is not a diversion. It is uttered as part of conversation / discussion and is normal; it is about the context.

(s1): A2 finds straw man due to 'taking god out of schools'.

There is a slight misinterpretation in that god is not necessarily out of school by getting prayers out of the schools; nobody is deprived of her religion or belief. However, labelling this as a fallacy requires some contextual knowledge and since the misinterpretation is slight, we can leave the sentence as non-fallacious.

(s2): A2 finds appeal to emotion due to 'religion in jails'.

This labelling is due to lack of contextual knowledge. What the writer talks about is some past event so it is not an emotional narration.

(s4): A2 finds false cause due to 'students learn more'.

This is not a false cause since the improvement in learning is not attributed to only one reason.

(s5): A2 finds ad hominem and red herring due to 'they could do'.

There is a signal of these two fallacies; however, it is not directly aimed at the opponent and is the writer's piece of opinion rather than a substantial part of the overall argument.

(s7): A2 finds ad hominem due to 'you would send'.

The writer is using a persuasive language and assuming that the reason not to commit to private schools is lack of money. This is a slight attack and is directly related to the argument so the sentence has the fallacy of ad hominem.

(s10): A2 finds straw man due to writer's mentioning of some monetary policies.

When it is considered with the previous two sentences, we can say it is straw man since the writer continues misinterpreting the objection to private schools as a problem of money. However, this sentence alone has red herring, the writer shifts to another topic.

doc#4068

premise: S0I have sent my son to private schools so that he could avoid the negative elements of society. S1I have been advised to put my money in private schools, or else save it for attorneys later. S2It would be silly to sacrifice him to public schools to make them better. S3It wouldn't work anyhow.

A1 finds no fallacies and A2 finds several fallacies.

(s0): A2 finds straw man due to 'negative elements of society'.

The writer is not misinterpreting any other claim but giving her reasons about her choice, which is plausible. The sentence does not have an apparent fallacy.

(s1): A2 finds appeal to consequences due to 'or else'.

The writer presents that the choosing public schools always ends up in courts so the sentence is fallacious.

(s2): A2 finds appeal to emotions, straw man and red herring due to the words 'silly' and 'sacrifice'.

The style is in the tone of normal daily conversation so it does not have fallacy.

(s3): A2 finds hasty generalisation due to 'it would not work'.

The writer tells her opinion, it is not a statement of generalisation. The sentence is not fallacious.

3.2 commentset2

Another 20 documents that contain user comments are selected from our total collection (341 texts). The set has a total of 126 sentences while the average number of sentences per document in the set is 6. The same annotators annotated each sentence of each document in this set and this time in a much shorter time, as well as resulting in agreement.

Improvement in agreement can be attributed to the previous verbal and written discussions about disagreements and working on the guidelines.

3.2.1 Agreement results

In Table 4 we present our agreement results.

distance metric	agreement value
MA SI distance metric	0.1967
set distance metric	0.1969
nominal distance metric	0.1973

Table 4: Agreement results based on Krippendorff’s alpha of the annotations in commentset2

Nominal distance gave slightly a better result than the other set distance metrics most probably because it disregards multi-labels.

3.2.2 Label-based analysis

We conduct this study to understand the capturability of our fallacy types. This will also help us distill our guidelines / set of types better.

Overall frequency of the labels Below in Table 5 we present the number of times any of the annotators used the corresponding label (or set of labels) for a sentence. As we have 126 sentences and two annotators, there is a total of 252 annotations.

labelid	label name / fallacy type	# of times used
1	argumentum ad hominem	1
2	biased sample	7
3	composition / division	1
5	false cause	2
6	begging the question	2
8	appeal to consequences	3
10	appeal to force	1
12	appeal to emotion	9
14	straw man	3
15	red herring	1
16	appeal to sarcasm / ridiculising	6
na	no fallacies	214
1, 14	ad hominem + straw man	1
14, 15	straw man + red herring	1

Table 5: Frequency of the labels in the annotations in commentset2

The most chosen labelling is non-fallacious and the most chosen fallacy type is appeal to emotion. As for the second mostly used fallacy type, namely biased sample, we can say that it is usually used for a span of multiple sentences since the writers usually tell a story through a multiple of sentences and draw a conclusion out of this segment so this is one of the reasons biased sample occurred relatively high, 7 times. Straw man fallacy co-occurred with red herring and ad hominem fallacies, respectively. They are quite interrelated similar fallacies and there is not a clear-cut distinction among them.

Below we list the fallacy types that did not occur in our annotations:

- (4) false analogy: This fallacy is usually captured in the scope of false cause or appeal to consequences. A remarkably vigorous annotation is needed to look for it, as well as broader and relatively well-written texts.
- (7) argumentum ad ignorantiam: Not detected though not hard to capture.
- (9) sunk costs: This fallacy is usually captured in the scope of appeal to consequences. A remarkably vigorous annotation is needed to look for it, as well as broader and relatively well-written texts.

- (11) argumentum ad populum: Not detected though not hard to capture.
- (13) appeal to false authority: Not detected though not hard to capture. (usually in the form of not-quoted resources or authority validations)

Agreement rank of the labels Below in Figure 2 we present the coincidence matrix of our annotations in this set.

	na	2	14	12	8	14, 15	1	5	16	10	6	15	3	1, 14	Σ
na	380	12	2	8	4			4	8	2	2	2	2	2	428
2	12		2												14
14	2	2					2								6
12	8					2									18
8	4										2				6
14, 15				2											2
1			2												2
5	4														4
16	8														12
10	2														2
6	2				2										4
15	2														2
3	2														2
1, 14	2														2
Σ	428	14	6	18	6	2	2	4	12	2	4	2	2	2	504

Figure 2: Contingency matrix.

Here we examine some of the cases one by one.

- **no-fallacies:** 95 sentences were found non-fallacious by both annotators. It appears the easiest pattern to capture due mostly to the type of the texts (stories continue throughout sentences, which can well be fallacious when that chunk considered; informal tone omitted for fallacies, knowledge of the domain and the discussions etc.) and the agreed inherent ambiguities (linguistic or domain-related).
- **appeal-to-emotion:** Two sentences were agreed that it contains the fallacy of appeal to emotion. They contain explicit emotive words / subjective expressions: (doc#163-s9: “That the Commonwealth of Virginia has such a rich tradition of the education of young people and allows this travesty is shameful.”; doc#2616-s1: “It was shitty prior to her birth or mine?”). This label co-occurred mostly with no-fallacies and (straw man, red herring) pair. This label is the second mostly used label.
- **appeal-to-sarcasm:** One sentence is agreed on this type, which is apparently sarcastic. (doc#1546-s2: “If you want to chat with your invisible, supernatural friend do so off school property and on your own time.”). It co-occurred with no-fallacies, only.
- **straw-man:** It is never agreed on. It is confused/co-occurred with biased sample and ad hominem. The pair it is a member of, namely (straw man, red herring), is confused with appeal to emotion. In addition, this pair is never confused with no-fallacies.
- **ad hominem:** This is one of the two labels that were not confused with no-fallacies. And it was confused with straw man (see disagreement study doc#902-s2).

3.2.3 Disagreement study of the annotations in commentset2

We present the analysis of disagreements on our annotations of commentset2, whose agreement values we presented in the previous section.

doc#163

S0Thank you for bringing this tragedy to light. backing: S1I am a Christian, an educator, a student and a parent and I have seen too many children like the Powells. S2As an admissions officer, we had applicants whose "record keeping" consisted of sending boxes full of paper for our office to review as part of the application. premise: S3If their students did get an interview, which was rare, they didn't have the social skills to survive the first round.

premise: S4I personally am acquainted with four families who are home schooling their large families. S5All four have no intention of book-schooling their daughters past age 13 as they need to learn :homemaking skills". S6One of the girls, who has not been taught for two years, could be Josh Powell's twin. S7She is intelligent and desperate to learn, but her parents won't allow it. S8It is heartbreaking.

S9That the Commonwealth of Virginia has such a rich tradition of the education of young people and allows this travesty is shameful. S10All of us, no matter our religious beliefs, need to pray that the law changes before more smart children are left behind.

Both annotators agree on appeal to emotion in one sentence and both find some other disjoint fallacies.

(s1-s7): A1 finds biased sample due to the personal story told, though indicating being unsure.

It is not fallacy actually, the writer only tells an experience and does not generalise it; on the contrary, she expresses her opinion about the very incident.

(s7): A2 finds straw man due 'parents won't allow her to learn'.

The writer interprets homeschooling as 'not allowing to learn'; this is exaggeration and manipulative.

(s10): A1 finds appeal to consequences due to 'before children left behind'.

The writer is supporting her claim by pointing to an unpleasant consequence which may not necessarily follow from the current situation (law is not changed).

doc#291

S0Adults have every right to make decisions for themselves for which they later suffer. S1. premise: S2Consciously depriving a child of an adequate education solely because "father knows best," or thinks he does, is tantamount to child abuse.

premise: S3When it comes to medical care, the courts often overrule parents who are irrationally threatening a child's life due to mistaken religious beliefs. S4If a teenager told doctors, "I want this treatment" and the parents refused to give consent, what would a court do? premise: S5An adequate education is vitally important in a child's life. claim: S6Parents should not be allowed to withhold it.

S7How many young people have the intelligence, drive and courage to do for themselves what this young man did? S8He is remarkable.because "father knows best," or thinks he does, is tantamount to child abuse.

Both annotators mark the same two sentences but with different labels and A2 finds one more.

(s2): A1 finds appeal to emotion due to the writer's comparison of homeschooling with child abuse while A2 finds it red herring. A2 additionally finds straw man in this sentence due to 'fathers know the best'.

Both A1 and A2 are right; indeed, diversive and emotional fallacies are usually intertwined, very similar. The writer misinterprets people's choice about education, attributing it to the fathers' decision, which is an overinterpretation.

(s7): A2 finds appeal to emotion due to the question with a comparison.

The writer appeals to the courageous feelings of the audience and it feels persuasive. There is a slight fallacy and it is actually considerable as some part of custom conversation.

doc#902

S0what like making sure you are fit to teach a child responsibly.. S1I'm sure you are, but there are plenty that aren't. premise: S2Or you could stop being so selfish to the child and send it out into the world so it can learn the importance of society and how to ineract with it... an education is often only useful if you know how to work with people using it.

Both annotators find a sentence fallacious but disagree about the type.

(s2): A1 finds ad hominem and A2 finds straw man due to 'being selfish'.

It is actually an attack on the opponent, at the same time it is a diversion in that the author misinterprets the motivation of the opponent action as selfishness. In this sentence, ad hominem is more apparent than straw man.

doc#1064

S0@oommph - here you get to a sensitive subject - modern science contains hardly any proper science, just compare the exam papers from the past decade with those from the 70s and 80s. S1You're also making the mistake of directly comparing discrete subjects from school with a more broad-based approach used by home educators. premise: S2We teach our children how to learn, not merely how to pass tests. backing: S3My 8yr-old is already enjoying our sessions with basic physics and chemistry textbooks, books which are generally used at the start of secondary school. S4He's interested now, so we cover the subject now and he'll learn it much quicker because of that interest.

premise: S5Also consider that school is about 25% efficient, so of the five hours a day of nominal lesson time, you lose a fair bit in disruptions, admin, not understanding (or not listening to) the topic and having it explained again, finishing the exercise and waiting while it's explained again to someone else, etc. Home education doesn't take long holidays like school does, so if you work it out, the 250 useful hours of learning at school during a year is easily matched and exceeded by home educators who are doing low-key, low-pressure teaching all the time. backing: S6We've even done addition and subtraction exercises on a flight home from a holiday before now, and in a Parliamentary committee room one evening, so you can see we're all quite flexible about learning.

A1 finds no fallacies and A2 finds one.

(s0): A2 finds false cause due to the writer's presenting the poverty of science as the reason why exam papers are poor. *This is implicit though the annotator is right.*

doc#1260

backing: S0If Muslims are required by their religion to pray several (is it 6?) times a day, and this space is being allotted so the Muslim youth can practice their religion properly... why is this prayer only being conducted once per week? S1This timetable would seem in opposition of what their faith demands. S2I am convinced more satisfaction is gained from the fact that they will be allowed to halfheartedly practice their faith in a public school rather than actually practicing their faith properly an a place of worship.

premise: S3And boo hiss on any Canadian institution that would allow discrimination against gender to be practiced on tax-payer funded property.

A1 finds sarcasm and A2 finds no fallacies, though indicating that the text is weird.

(s0-s3): A1 finds appeal to sarcasm due to the writer's comments about islam.

It is not so explicit and indeed what he tells is true about islam so not necessarily sarcastic.

doc#1546

claim: SONO PRAYER IN SCHOOLS!...period.

premise: S1Our education system (as with our government) must remain secular and free from these silly distractions. S2If you want to chat with your invisible, supernatural friend do so off school property and on your own time.

Annotators once agree and once disagree.

(s0): A2 finds appeal to force due to exclamation and 'period'.

It feels an attempt to stop the discussion without any reason but with some shouting. The annotator interprets this as 'stop this claim, otherwise I stop it'. The flaw in this sentence can be begging the question due to being circular.

(s1): A1 finds begging the question.

This is just a statement connected to its context. The writer does not present her claim with the very claim as a reason.

(s2): Both A1 and A2 find sarcasm in this sentence.

They agree and they are right.

doc#2036

backing: S0I am going to be starting a single sex school but am not intimidated by this at all, as it was not just my parents choice but mine too, alot of people I know quite well go there and enjoy it tremendously. S1Its just a debate that has two extremely good arguments on both sides

Both annotators finds this text non-fallacious.

doc#2180

claim: S0Personally i'd go co-ed.

backing: S1As someone who went to a same sex school for 8 years, I found it lacked the diversity you get in a co-ed environment. premise: S2I found the attitude and behaviour of students in the co ed school to be better, and i attribute that to the influence of the opposite sex. premise: S3There's no doubt boys behave a little different when girls are watching, and i also found boys were quite good at limiting the bitchyness girls are renowned for. S4So both kept one another in line, and made for a more positive and dynamic environment.

premise: S5I also think there's a few extra life lessons and skills children can learn at co ed schools. S6Dating, relationships, interacting with the opposite sex, i think children at co ed schools tend to have a far better grasp of these skills then students who've only attended same sex schools.

Both annotators finds this text non-fallacious.

doc#2616

S0So I am suppose to sacrifice my child education, for the sake of the countries? S1It was shitty prior to her birth or mine? premise: S2I am a single mother and I make less than most people, and I am sending my child to a private school. S3So I guess I am a bad person!

Annotators agree on one fallacy and A1 finds some more.

(s0): A1 finds appeal to emotion due to sacrifice.

This comment is an answer to another writer who claims choosing public schools causes improvement. This writer interprets that suggestion as sacrificing the children, which signals straw man. Using the affective verb 'sacrifice' is an attempt to appeal to emotions, on the other hand. We can say the sentence contains both of these fallacies.

doc#2823

claim: S0I agree - Kids can do great in the public school system and parents DO need to be involved. S1The more people leave, the worse its going to become. premise: S2The public school system lets them deal with real life too, unfortunate that it may be but that is what's out there in college and the work force too. premise: S3There are still great teachers in the public schools - lets stand behind them.

Both annotators find one sentence fallacious but disagree about the type.

(s1): A1 finds appeal to consequences due to the writer's attempt to dissuade the public school leavers by showing the anticipated result in that case. A2 finds begging the question to the writer's not showing any reason why schools get worse upon people's leave while the reason for leaving is the undesired state of the schools.

What the writer says about the consequence can be true so it is not necessarily manipulative. And it is not circular, the writer is only showing the effect of an action.

doc#3313

claim: S0My child went to a private all boys school and it was the best thing I ever did! premise: S1He went in a boy and came out a god-loving man of character! premise: S2I believe a more disciplined school in the formative teenage years pays off when the child matures. S3In public school, the parents nor the teachers have too much say in the government run institutions!

A1 finds no fallacies and A2 finds one.

(s3): A2 finds red herring due to the writer's mention of the right to speak in governmental places.

What the writer says can be true and can well be a reason for not choosing public schools. It is not necessarily diversive.

doc#3504

S0When you have an entire citizenry that has been raised to hate unto death their families, their neighbors and each other, then anybody with a brain in their heads is deemed "different" – and not in the good way. S1Our entire society is based upon hate. S2Our government is based upon hate. premise: S3Our entire public school system is nothing but a battleground controlled by hate and a hatred that is enforced by bullies. S4Knowledge is power. S5Knowledge is the only thing that will save you. claim: S6The best education is the only route to go; and it isn't in a public school system.

A1 finds no fallacies and A2 finds two fallacies.

(s0): A2 finds composition / division fallacy due to his defining entire citizenry living in hate.

There might be some people in that state, they might be many indeed; nonetheless, it shouldn't be attributed to the whole citizens and the concept of citizenry. The sentence is fallacious.

(s3): A2 finds false cause due to 'public schools controlled by hate'.

There might be other reasons and this might be the author's opinion as well. However, there is an apparent causal flaw.

doc#3919

claim: S0Parents have the right to send their children to private schools. S1They pay for it in addition to paying for the public schools in their taxes. S2So they do invest in public schools even if their child does not attend them. S3Stay out of people's personal business!

A1 finds no fallacies and A2 finds one sentence fallacious with two types of fallacies.

(s3): A2 finds straw man due to 'personal business' and ad hominem due to 'stay out'.

The writer misinterprets a family's choosing the private schools as a personal issue. And the people who are against this choice are attacked by warning them to stay out.

doc#5059

S0Why is it called redshirting, I wonder?

S1My son was born in December (cut-off Dec.31) and I have been a little worried for him because they would want him to start JK when he's only 3! S2That's still a baby! S3But I think I'll just keep him with me until SK and then let him go. S4I'm not pleased with the full-day kindergarten coming in, but I'll worry about that when it gets closer. S5(he's 2 now)

I thought of holding him back, but my husband was always bored in school and so if he's smart like his daddy it might actually work out very well for him to be a bit younger than the other kids. S6(physically he's an ox, takes after my side of the family!) S7Besides, you need to think of the kid's future. premise: S8If all the others are graduating high school at 17, why does he need to be there until he's 18? S9His whole life is delayed by a year. S10That's not a favour, in my opinion.

S11I'll make sure to remind the teacher of his tender years if he starts getting poor marks for "scissoring skills". S12:P (surely you don't get marks in Kindergarten? S13I have no idea.)

A1 finds no fallacies and A2 finds one.

(s10): A2 finds appeal to emotion due to 'favour'.

There is a hidden appeal and it may just be an innate feeling about the children. There is a negation of the opponent claim with this emotive word, indeed it is diversive, but it is not apparent. The sentence is not apparently fallacy.

doc#5064

S0In Chicago, Illinois (USA), the school district instituted a policy that required 6 YOs to be placed in 1st grade if their parents held them back from Kindergarten. S1Redshirting has become such an issue there that the school district felt they had no choice.

premise: S2I can certainly understand it if the parents feel the kid isn't ready, but what I'm hearing a lot of in our area is parents (particularly of boys) doing it because they feel it will make their kid the "class leader" or the "best" in sports (which is ridiculous IMHO). S3One of the girls I went to high school with went redshirted two of her three boys because of the sports issue alone (apparently, she didn't know she had the option with the first one but would have redshirted his too if she had known). S4She claims it makes a "huge" difference for sports. premise: S5Personally, I'm not sending my kids to school to play sports. S6I'm sending them there to learn. S7Plus, from my own sports experience, age has little to do with sports once you hit the high school level anyway – I played Varsity Volleyball and Varsity Track as a freshman (when I was 14). S8Most of the team was made up of seniors and juniors (who were 3 to 4 years older than me). S9And really, how can you as a parent feel proud of your child and feel that he is "better" at sports than his classmates when his classmates are 2 years younger than him. S10Seems a tad delusional to me.

S11For us, we are having the opposite issue – we are sending our 4 YO twins to kindergarten early because they are academically, socially and physically ready. S12They miss the cut off by three months here, so it's been an issue finding a placement for them. S13So all this redshirting means that they are likely to have kids who are 2 years older than them in the same class as them by first grade.

Both annotators find no fallacies.

doc#5291

backing: S0My son turned 5 one week before the cut-off. S1He is in a full day Kindergarten and seems to be doing well, considering he is the youngest in the class. S2Last year, his pre-school teacher recommended that I hold him back and keep him in Pre-K one more year. S3Pre-K was only 3 hours long, while Kindergarten was a full day. S4My husband and I agreed, that if our son has to be held back, it will be when we're sure he's not keeping up with the kids and Pre-K is not that time. premise: S5He would have been bored going over the same things and not getting challenged. backing: S6He has had no problems so far in Kindergarten. claim: S7I'm happy with my choice and it was comforting to read this article of support for my decision.

Both annotators find no fallacies.

doc#5330

S0Redshirting a trend? S1This is insane! premise: S2These kids need to be prepared for the 21st century global economy by being enrolled in a local second language immersion kindergarten as soon as they can enroll. S3This is because language acquisition theories tell us that, "there is a window of opportunity in which the child learns that first language normally. S4After this period, the brain becomes slowly less plastic and by the time the child reaches adolescence, the brain cannot develop "richly and normally any real cognitive system, including language." http://www.cal.org/earlylang/benefits/research_notes.html. As Larry The Cable Guy would say: "Only in America!"

A1 finds no fallacies and A2 finds several fallacies.

(s1): A2 finds appeal to emotion due to 'insane'.

This sentence contains argumentum ad hominem rather than appealing to emotions.

(s2): A2 finds appeal to consequences due to 'if not, then not prepared for 21st cc'.

The writer mentions very broad and unrealistic possible results of the opposite action. The sentence is fallacious.

(s4): A2 finds straw man due to the writer's mention of language learning difficulties at some age.

The writer is exaggerating. The latency is not for ages and it is not very likely, except for particular cases, that the children will be affected for their language acquisition skills due to attending the school a few years late.

doc#5345

backing: S0We kept our middle child back one year. S1She turned 5 in early September, but our thought was emotionally, she wasn't ready, so she started at 6. S2We always did a lot of reading and exploring the world we live in, so keeping her mind busy wasn't a problem. S3She is now a lawyer, who graduated in 2011, president of the law school bar association, hired right out of school and doing very well. S4What struck me throughout her school years was her total confidence as she went through school. S5She always felt she could succeed and did.

Both annotators find the text non-fallacious.

doc#5351

backing: S0I can attest to the idea that things even out by 3rd grade. S1My daughter was 4 when she started kindergarten, turned 5 in late Sept. She thrived socially, but was definitely "middle of the pack" academically till the end of 2nd grade, when something kicked in, and she started caring more about doing well in school. S2Now she's in 7th grade, and still at the top of her class. premise: S3I worry some about her going off to college at age 17, but the idea of a gap year at 17 to engage in a rewarding volunteer experience sounds better to me than an extra year spent in preschool would have been.

Both annotators find no fallacies.

doc#5418

S0He sounds ready. S1What IS the cut-off date? S2Ours is Sept. 30th, and no one "red shirts". S3There is about 5 cases a year for the whole province of Quebec, mostly very immature children. claim: S4Three of my girls were born in August, but there is no question of holding them back. backing: S5Everybody in my family (my siblings, husband's sibling)skipped a year in school i.e. moved forward faster.

Both annotators find no fallacies.

3.3 Decreasing the number of categories / Merging the categories

In our initial annotation studies in commentset1 and commentset2, we had 20 and 16 categories (labels), respectively. In the course of our label-based and disagreement analyses, it appeared that there are inter-related, very similar categories. Indeed, we have found that having four categories of fallacies, namely 1) causal, 2) consequence-related, 3) diversion-targeting, 4) emotional/sarcastic, can relieve the annotation process on the side of the annotator and increase agreement, as well as being realistic and consistent.

In this regard, for each annotation set, we joined the related categories in these four groups and re-calculated the agreement scores.

3.3.1 Agreement scores of commentset1 after merging categories

In Table 6 we present the new labels after locating them in a suitable group.

previous (labelid) label name /	new (labelid) label name
(1) argumentum ad hominem	(3) diversion
(2) hasty generalisation / specification	(1) causal
(3) biased sample	(1) causal
(4) anecdotal arguments	(2) consequence
(5) composition / division	(1) causal
(6) false analogy	(1) causal
(7) false cause	(1) causal
(8) begging the question	(1) causal
(9) argumentum ad ignorantiam	(1) causal
(10) appeal to consequences	(2) consequence
(11) slippery slope	(2) consequence
(12) sunk costs	(2) consequence
(13) appeal to force	(2) consequence
(14) appeal to popularity	(1) causal
(15) appeal to emotion	(4) emotion
(16) appeal to false authority	(1) causal
(17) straw man	(3) diversion
(18) red herring	(3) diversion
(19) appeal to sarcasm / ridiculising	(4) emotion
(20) trivialising / trivial objections	(3) diversion

Table 6: Fallacy types (category labels) used in annotating commentset1 mapped to the super-groups

According to this mapping, we reformed the annotation table and re-calculated the agreement scores using again Krippendorff's alpha with three different distance metrics as there were still multiple labellings for an item. In Table 7, we present comparative results.

distance metric	agreement value	
	20 categories	4 categories
MASI distance metric	-0.0233	-0.0342
set distance metric	-0.0234	-0.0301
nominal distance metric	-0.0309	-0.0150

Table 7: Agreement results based on Krippendorff's alpha of the annotations in commentset1; with 20 categories and with 4 main categories, separately.

In the Table 7 we see that agreement value remarkably decreased upon having 4 categories when we used the set distance metrics (MASI and set distance, first two rows). However, when we use nominal distance metric which omits multiple labellings, agreement has risen considerably. We can say that generally, multi-label cases are the source of disagreements. In the single-label cases, having main categories (only 4 cats) seems to have resolved many of the disagreements. We can also say that our inference that labels are inter-related and similar is validated by these results.

3.3.2 Agreement scores of commentset2 after merging categories

In Table 8 we present the new labels after locating them in a suitable group.

previous (labelid) label name /	new (labelid) label name
(1) argumentum ad hominem	(3) diversion
(2) biased sample	(1) causal
(3) composition / division	(1) causal
(4) false analogy	(1) causal
(5) false cause	(1) causal
(6) begging the question	(1) causal
(7) argumentum ad ignorantiam	(1) causal
(8) appeal to consequences	(2) consequence
(9) sunk costs	(2) consequence
(10) appeal to force	(2) consequence
(11) appeal to popularity	(1) causal
(12) appeal to emotion	(4) emotion
(13) appeal to false authority	(1) causal
(14) straw man	(3) diversion
(15) red herring	(3) diversion
(16) appeal to sarcasm / ridiculising	(4) emotion

Table 8: Fallacy types (category labels) used in annotating commentset2 mapped to the super-groups

According to this mapping, we reformed the annotation table and re-calculated the agreement scores using again Krippendorff's alpha with the nominal distance metric (the other set distance metrics give the same result). We should note that upon merging the categories, multiple labellings disappeared. So in this new case we have only single labels per item. In Table 9, we present the results for both cases.

distance metric	agreement value	
	16 categories	4 categories
MASI distance metric	0.1967	0.2153
set distance metric	0.1969	0.2153
nominal distance metric	0.1973	0.2153

Table 9: Agreement results based on Krippendorff's alpha of the annotations in commentset2; with 16 categories and with 4 main categories, separately.

Since the distance metrics are not the same (not applicable to one another), we cannot make a direct comparison. However, we can say that agreement is slightly better with less number of categories.

In Table 10 we present the contingency matrix of this setting.

	na	1	3	4	2	Σ
na	380.000	20.000	6.000	16.000	6.000	428.000
1	20.000		2.000		2.000	24.000
3	6.000	2.000	4.000	2.000		14.000
4	16.000		2.000	12.000		30.000
2	6.000	2.000				8.000
Σ	428.000	24.000	14.000	30.000	8.000	504.000

Table 10: Contingency matrix.

From Table 10 we see:

- The most agreed category is no-fallacies (na). This is followed by the category emotion (4), agreed in 3 sentences and then diversion, agreed in one sentence.
- No-fallacies is confused (mistaken) mostly with causal fallacies (in 10 sentences) and diversion (in 8 sentences) .
- Consequence-related fallacies used very few times (4 sentences by any annotator) with no agreement achieved.

3.4 blogpostset

10 documents are selected from the same collection (341 texts), which were previously annotated for argument components. This time the texts blog posts or articles, which are longer, relatively more coherent and topical.

3.4.1 Disagreement analysis of annotations in articles and blogposts

We apply a disagreement study of the same annotators' annotations on blogpostset.

doc#414

A1 finds no fallacy "in the sense what we are looking for" and A2 finds fallacies in three sentences.

(s7): "[H]e has a right to educate his children any way he sees fit without the government interfering" : A2 says false cause because of "any way he sees".

However, this is just an opinion.

(s21, s22): "As Clarence Powell himself says in the Post article: "A good education is not an option. It's essential": A2 says this contains lack of evidence due to defining the object as essential.

It is not explicit, though gives a clue of fallaciousness. There is no sufficient evidence for marking it fallacious. A1 is right.

(s25): "[“]If you're a loving parent, the last thing you want to do is create a situation where your children are limited or hindered.”: A2 finds appeal to emotion due to 'loving parents' since there is a persuasive language and not hindering children should not be only about being parent and even a loving parent.

This is detailed and actually the sentence is not very argumentative or relevant to the argument at hand at all. A1 is right, it is not fallacious.

doc#850

A1 finds two fallacies in two different segments respectively, finding one of them weak. A2 finds fallacies in 9 sentences.

(s5-s9): A1 senses appeal to consequences but says it is not fallacious.

It does not signal so fallacious, yes.

(s10): A2 finds lack of evidence (appeal to false authority indeed) in the next sentence. She marked it because this fallacy is actually what the writer found and reported in some quoted resource.

There is literally no problem. This is not a direct fallacy marking, it is a note.

(s12): “The education officers who liaise with home-educators are trained teachers, not social workers, and so are not trained in the highly specialised field of child abuse.” A2 finds ridiculising due to ‘highly specialised field of child abuse’. *Considering the context and the writer’s side to refute the claims of child abuse in homeschooling, the writer is ridiculising and making use of sarcasm. However, this sarcasm is not related to the reasoning so there is no fallacy.*

(s13): “Under the new regulations though, they would have the legal right to enter my home and interview my child alone in a room to “check” for signs of abuse, not because of any good reason to suspect abuse is taking place, but simply because I have chosen to home educate.” : A2 finds straw man due to ‘not for any good reasons’.
*There is a straw man but here the argument is not directly in the main argument, it is a different argument alone. The writer is against the authorities’ checking home. The authority applies this to homeschoolers to check if there is any abuse. And the writer misinterprets this and understands as if he is being distracted *just* because he homeschools, kind of essentialism. Again, if we are to consider non-argument components and non-annotated sentences, then yes, there is straw man; otherwise, we omit this. Same holds for the sentences s14, s15. Indeed, A1 also thinks s14 contains some pleading/appeal in an implicit way.*

(s17): “As with so much legislation that the government has managed to get through parliament under the emotive banner of “child welfare”, this proposed legislation peddles the idea that children can only be dealt with by professionals.” : A2 finds false analogy due to ‘as with other legislations’.
This is implicitly and somewhat (in terms of linguistic patterns) formally false analogy. He makes a prediction based on his observation of the other related legislations. Since these legislations are on the same topic, then their consequences will be the same. Formally it is flawed but real world knowledge may render this sentence non-fallacious.

(s18, s19): “More and more parents are choosing to home educate because the school system fails their children. There is widespread complaint among parents that children are tested too young and too often, teaching them only to pass tests so that civil servants can compile ultimately meaningless statistics.” : A2 finds lack of evidence / false cause in both due to ‘more and more’ and ‘widespread complaint’, respectively.
The writer tells this without evidence though what he says might be true and a general, widely known situation in his environment. So we can say it is not fallacious upon context. A1 is probably right.

(s23): “But typically, the state tends to make the worst parent of all, and I don’t need its kinds of lessons.” : A2 finds straw man due to ‘state tends to make worst parent’.
The topic is homeschooling with some mentions to state policies. However, here what the writer says is completely different and a kind of blame, prone to open a new discussion with a different claim. It is like misinterpreting states’ policies, considering the opponent is the state. Yes, the sentence has straw man.

doc#1045

A1 finds no fallacy except an implicit one in the last sentence and A2 marks two sentences as fallacious.

(s3): “It seems that Khyra’s last hope was that the local authority’s home education support team might intervene once her mother had withdrawn her from school.” : A2 thinks there is red herring and ad hominem due to ‘her last hope’.
There is no fallacy, there must have been a misunderstanding.

(s24): “And if the answer to that is ‘Maybe, but those children are still alive’, then consider this: an estimated 16 children per year commit suicide because of problems at school.” : A1 thinks there is an appeal but says it can just be an evidence and A2 finds biased sample due to the report about children suicides.
There is a missing evidence and the writer doesn’t tell where these schools belong to, which is manipulating. Indeed, there is rather red herring since the issue is shifted to children’s severe problems. It can be regarded as an appeal to emotion.

doc#1666

A1 marks three sentences fallacious, one being uncertain. A2 marks seven sentences. There is an agreement between the two on one sentence.

(s3, s7): “What is this ‘spontaneous eruption of religiosity’ that FFRF finds to be so threatening to students at public schools?”, “Yet there are many people and groups, like FFRF, that find the peaceful prayers of this gathering of Christian students to be threatening.” : A1 finds the 2nd one (s7) emotive though not necessarily fallacious. A2 finds straw man in both due to the expression ‘threatening’.

According to what the writer reports about the opponents' (ffrf) claims, there is no threat reported, what the opponent says is just very negative. It sounds emotive but fallacious because of this, as A2 says. There is a mild straw man fallacy.

(s8, s9): "No one is being forced to participate in the event, nor is it interfering with class time. So what is the threat?" : A2 finds straw man due to the mention of threat.

Similar to the previous (s3, s7) explanation.

(s16): "But if prayer really works, then that group of students calling out to God around the flagpole becomes a threat to those who deny the God's existence or His sovereignty over this world" : Both annotators find appeal to emotion and A2 also finds straw man.

The annotators agree and the appeal is evident. Continuation of the previous two cases, with the interpretation word 'threat'.

(s17, s18): "If prayer really works, then it is a threat to those who would seek to corrupt a generation of students into embracing a godless worldview. If prayer really works, then students might turn to God and come to know His love and forgiveness." : A1 finds appeal to emotion in the sense of faith.

There is a mild appeal to emotion, not directly related to the argument. The sentences are valid in their own right (according to the rules of faith and religion though none is rational in real.).

(s23): "An impotent prayer is no threat to anyone." : A2 finds straw man due to threat.

Same as the previous cases.

(s24): "But an effective prayer to a God who hears ... well, I guess some might find that a little threatening." A2 finds ridiculising / sarcasm.

Not really and not successful in the intended sarcasm and not relevant to the argument.

doc#1671

Annotators agree on two segments of text. A2 finds fallacious some 7 sentences/parts more.

(s2, s3): "Humanism assumes that the supernatural does not exist and that reality must be discovered purely from man's reasoning.", "Our public school system has been operating on this foundation since prayer was removed in 1962." : A2 finds false cause due to the writer's interpretation of humanism-religion relation.

This might be false cause but we need to know whether the prayers were abandoned in the schools on the grounds that the policies aimed at abandoning humanism as well.

(s13): "Most Americans are in favor of prayer in public school because they realize that there is something to be gained from the open acknowledgement of God's existence." : A2 finds lack of evidence due to 'most americans'.

There is a signal of false cause but what the writer says can be true in the context.

(s14, s15): "Morals can only be built on the basis that a moral source-God-exists. Once acknowledgement of God's existence is removed, there is no absolute basis for morals, and a free society drifts toward anarchy." : A2 finds appeal to false authority due to the given god-morality relation.

The writer talks about some absolute things which he lean on authority, which is false. The sentences are fallacious.

(s18): "What a hypocritical message we send to our children when we allow laws to be passed which limit the free expression of speech in public by banning public prayer in classrooms." : Both annotators find red herring; A2 finds also straw man. (A2 finds red herring in the following sentence s19).

There is not straw man, the writer doesn't attack the opponent's claim, and the re-interpretation is about the consequences of the action not the claim itself. Relating religious acts at schools to free speech is red herring, it is a far different, irrelevant issue.

(s21-s24): "Our nation was founded on the freedom to publicly acknowledge God's existence." ... : Both annotators find it problematic but couldn't mark a specific fallacy. A2 says lack of evidence additionally.

Requires world-knowledge. Maybe he is telling something true however irrational it is.

(s26): "As with all truths, this is still applicable today." : A2 finds appeal to emotions and red herring considering the previous sentences (there the author represents atheists as evil and believers as good).

This is his opinion, the emotional appeal is not directed towards the argument. Good-evil comparison can be red herring but

slightly. We can skip this part, it is very outside of the topic.

doc#2096

A1 finds the text non-fallacious and A2 finds one fallacy in a sentence.

(s1-rebuttal): “[A]lthough data would indicate that single sex girls’ schools might produce the best results, it would seem to be that there is more to this debate than just academic results.” : A2 finds lack of evidence (appeal to false authority) due to ‘data would suggest’.

The writer makes a prediction, possibly to increase credibility, and leans on a false /non-existent authority, the data that is not present. It is fallacious.

doc#2163

A1 finds non-fallacious and A2 remarks upon an issue about citations, that is the question as to whether we should investigate the reliability of citations on fallacy study, in the sense of appealing to false authorities.

This is detailed and excessive. The text has no fallacies.

doc#2230

The annotators agree on some 10 sentences about the fallacy of appeal to consequences. There are disjoint markings, thus disagreement again.

(s12): “I know that seems like such a minor gripe, but trust me, you don’t want to be surrounded by the bullshit girls can concoct amongst each other when there are no boys to ‘restrict’ certain behaviours.” : A2 finds appeal to emotion and authority due to ‘trust me’.

This is not fallacious. The expression stems from the informal tone of the style, not really persuasive and directed toward the argument. An experience is narrated, which can very well be true in its own right.

(s15): “I actually do feel like co-ed schools develop a girl better socially and emotionally than all-girls schools, just from the mere fact that a girl will get to interact with boys on a daily basis.” : A2 finds correlation-cause mix / false cause due to ‘just from the mere fact’.

This reason might not be the mere factor; there might be other reasons. It is an important issue and cannot be explained by only one reason. This is a fallacy of false cause.

(s16): “Another thing that I feel parents should be aware of: I personally found that girls who attend all-girls school will go out more to actually meet boys than girls in co-ed schools.” A2 marks lack of evidence due to ‘i personally found that’ and A1 finds appeal to consequences.

It is an opinion and does not strictly needs an evidence so there is no fallacy in that sense. The consequences that the writer mentions for the arguments follows up in the coming sentences so this sentence alone is not fallacious.

(s16-s27): A1 marks this range as containing appeal to consequences and A2 also finds appeal to consequences but in range s20-s27.

The annotators agree except the range, though the consequence story starts in s16, A1 is right.

(s26): “Girls LIKE male attention.” : A2 finds appeal to emotion.

It is an opinion and not directly about emotion. It might be the false cause of its upcoming claim but not evident.

(s28-s32): “The girls at my all-girls school got drugs from the boys they knew (all outside of school, of course, as it was an all-girls school). I didn’t know one person who did drugs at my co-ed high school, or know anyone who had access to drugs. But a friend in my all-girls school showed me some drugs IN school.” : A2 finds red herring due to the mention of drugs.

It also has appeal to consequences in addition to red herring. A1 might have missed.

doc#4382

Both annotators agree the text doesn’t have any fallacies.

doc#4581

One marking corresponds for both annotators though their labels do not match.

(s0-s2): “You are a bad person if you send your children to private school. Not bad like murderer bad—but bad like ruining-one-of-our-nation’s-most-essential-institutions-in-order-to-get-what’s-best-for-your-kid bad. So, pretty bad.” : A1 finds appeal to emotions as well as ad hominem and A2 finds false cause, red herring and hasty generalisation.

Rather than emotive appeal, there is ad hominem and it is the sum of the fallacies that A1 found. Disagreement most probably stems from A1’s detail-seeking marking.

(s14, s15): “Your local school stinks but you don’t send your child there? Then its badness is just something you deplore in the abstract.” : A2 labels begging the question.

The author shows a reason to a situation/consequence by mentioning one effect of the consequence, circular.

(s16, s17): “Your local school stinks and you do send your child there? I bet you are going to do everything within your power to make it better.” : A2 finds appeal to emotion due to ‘you do everything’.

It is an opinion, a call. If it said ‘you are a parent, you love your child, you will do everything so you will send your child to public school’, then we could say there is an appeal to emotion.

(s23): “There are a lot of reasons why bad people send their kids to private school.” : A2 finds red herring due to ‘bad people not choosing public schools’.

Not every provocation is a signal to red herring. This is, a continuation of s0.s3, is rather ad hominem.

(s34): “She may not learn as much or be as challenged, but take a deep breath and live with that.” : A2 finds the expression begging the question.

Considering the whole argument and its roots (generally good education), if there is no education or progress, then there is no need to discuss. The sentence is absurd rather than being circular.

(s37-s49): Personal story. A2 finds false cause, false analogy, hasty generalisation.

It is a story without much statements. No fallacy.

(s49): “I’m saying that I survived it, and so will your child, who must endure having no AP calculus so that in 25 years there will be AP calculus for all.” : A2 finds straw man since the writer diverts the topic from good education to survival.

It is implicit.

3.5 nytimes-debateposts

We collected and annotated 50 documents on about 7 different topics from the room for debates section of the newspaper new york times¹. The documents are user comments to some articles written on a controversial and current topic.

3.5.1 Disagreement study

We present some of the disagreement cases.

doc #15369835

1 I think you’re trying to say, so cleverly, that blacks aren’t really targeted by a racist system - they just really DO commit crimes all the time if law enforcement doesn’t ride their butts?

2 Have I got that right?

3 If they didn’t, for example, pull black drivers over disproportionately, God knows what blacks would get up to.

A1 finds the whole text sarcastic and A2 finds sarcasm only in s3. A1 is right.

doc #15370333

¹ <http://www.nytimes.com/roomfordebate>

-
- 1 Totally agree with the first paragraph.
 - 2 However, dropping out the euro and back into the drachma will create more problems than it solves.
 - 3 The Drachma would be quickly and greatly devaluated, inflation may ensue.
 - 4 Assume that on day 1 of leaving the euro the drachma is set at a 1:1 (parity).
 - 5 Very quickly the markets would devalue the drachma, as the value of any given currency is only as good as the government that issues it, and the governmental organizations that run the country and support the currency.
 - 6 "Ordinary" Greeks would have to be working harder and harder as the currency spirals downwards, particularly since Greece is still reliant on imports of raw materials, products and even services.
 - 7 The same structural problems of poor tax collection, heavy spending, early and large pensions would still be problem.
 - 8 When the drachma falls and drachma denominated assets and services are snapped up by foreigners, will the Greeks be screaming about economic imperialism, capitalist invasion or exploitation?

(s8): A2 finds both appeal to consequences and sarcasm. Actually the author is showing factual results, not presenting them as the sole consequences by skipping the main reason. The style is sarcastic in a way that leads to being fallacious.
doc #15478952

- 1 "Issues of ethics and religious liberty must be respected, especially when government funding is involved."
- 2 Thank you so much, Ms. Turner.
- 3 It's gratifying to know that government will respect my moral and ethical objections to war and, refusing to spend any more of my tax dollars on violent aggression against other nations and peoples, might yet thereby ensure peace in our time.
- 4 Poppycock!
- 5 You can't please all of the people all of the time.
- 6 The pronatalist anti-abortion, anti-contraception crowd just never passed eighth grade civics, apparently, and behaves as though it doesn't understand how representative republican government is supposed to work.
- 7 Sometimes, in the interest of best serving the interests of a majority, government will do things that might prove offensive to an extremist and vocal minority.
- 8 This does not mean that government should cease to function or to neglect the needs of the majority in an effort to placate minority interests.
- 9 When I look at teen pregnancy rates and infection rates for some STDs, my impression isn't that teens should have to "demand" birth control, but that Its distribution to any and all who want it ought to be mandatory!

(s6): A1 finds ad hominem. It is an apparent attack that is targeted against the opponent, for the purpose of supporting refutation.

3.5.2 Hard cases

There were a couple of cases that required real-world knowledge to detect the fallacy or a careful reading. We present them in what follows.

hard-case: doc #15351555

- 1 SteveRR, I am a minimalist in these matters.
- 2 I really want all Americans to confront what a real real civil war and the price of defeat.
- 3 The losers should have to stare at their failures frequently.
- 4 You can look at Lee and see a noble man, I do too.
- 5 I also see a loser.
- 6 When I was last in Memphis I made it a point to see Nathan Bedford Forrest who should have tried for a war crime.
- 7 I like his statue there because where Southerners see a warrior, there is actually a war criminal.
- 8 People need to learn both lessons.
- 9 Now my children, however, should not have to go to schools named after people like Vardaman who opposed all black education.

10 I am sure you would not send your children to Malcolm X University, or would you?
11 As for the monuments on the grounds where the common state does business, I'm for moving them, not destroying them.
12 If there were real Yankees left in America they would make some of this clearer.

A2 finds ad hominem in (s9). This requires real world knowledge and context relation with the previous sentences. It is an implicit attack but still apparently in use of argumentation.

hard-case: doc #15371571

1 Although leaving Euro would be difficult, that seems to be the only way left for Greece.
2 If they don't leave Euro, the ECB will never realize how important it is to maintain a country's infrastructure and tradition.
3 As a currency Euro is a badly conceived currency.
4 It has got the economic power but not the political power or control.
5 As everyone knows, economic power without political power is like putting the cart before the horse.
6 It frustrates everyone and ends up into a blame game.
7 Greece would be wise to leave Euro and implement some what harsher economic policies than what they have got.

A2 finds false analogy in (s5). This also requires real world knowledge. It is incorrect but this fault indirectly affects the validity of the argument.

A1 finds begging the question (s7). Knowing that the writer didn't propose concrete premises in the previous parts, it seems this sentence is only a rephrasal of the claim, causing a circle.

hard-case: doc #15472744

1 It's very telling when anti-choice mouthpieces like Turner find excuses to oppose a program like the Colorado initiative.
2 If these folks really believed that abortion is murder, then a program to reduce "murder" by 40% should be cause for joyous celebration.
3 When they start making excuses and changing the subject to "dignity", it tells us very clearly that they aren't actually concerned with "life" or "babies" at all.
4 What they are really upset about is people having sex.
5 They actually think that people enjoying sex is worse than murder.
6 How morally and mentally bankrupt.

A2 finds straw man in (s5). This requires the information about what the opponent said previously. Yes, the opponent says abortion is murder and sex causes abortion but does not say any of them (nor joy of sex) is worse than murder. The writer is involved in a misinterpretation.

hard-case: doc #15478399

1 Funny, I thought intercourse involved both a man and a woman.
2 But if we pay for the birth control it's because of "her decision"?
3 Well on that note, why should my insurance premiums go to pay towards men's Viagra?

A2 finds straw man in (s3). It is very implicit. This is another case that requires real world knowledge and information about the opponent's opinions particularly.

3.6 procon - online arguments collection - annotation1: annotating moderated texts written on debatable topics

We annotated 50 documents obtained from <http://procon.org>, the online collection of user generated arguments supporting or refuting a specific topic. We used webanno, the online annotation tool [Eckart de Castilho et al., 2014].

Not having exported the annotations, we manually controlled the agreement and it was 50%. Some examples of fallacious statements which were evident and agreed as well as the sentences which were confused to contain fallacies are added to the guidelines in Chapter 2 in their respective sections.

3.7 procon - online arguments collection - annotation2: annotating user comments to debatable topics

We annotated 56 documents from <http://procon.org>; this time each text is a piece of comment written by some users on the debatable topics presented at the site. These texts are not as professional as the previous procon texts and are not moderated and edited.

Agreement that was observed during the verbal curation process is better this time. The texts have more types of fallacies, do not have that many expertise information / knowledge. (compared to the previous case).

3.7.1 Hard cases

hard-case: doc #cID1090

Should people become vegetarian?

S1 Well I'm all for good health, but if human beings stopped 'eating mammals and fish' then what pray tell will happen with all the populations of cows, steers, pigs. lambs, chickens, buffalo, deer. elk and 'any' other four legged mammals and fish that human beings consume?

S2 What happens to all those animals and the 'impact' it would cause this planet.

S3 Not to mention the 'inferior' bunch of humans that would be produced from not having the core strength of proteins we as humans get from consuming animals and their meats.

A2 annotated S3 to have the fallacy of begging the question (circular) and the annotators couldn't easily agree on it. According to A2, the writer already assumes that eating meat and getting the benefits of meat has no other alternative and on its absence, one will suffer from producing 'inferiors'. Since this topical controversy contains the discussion of the alternatives of meat and its benefits, there is a jump from this point (assuming that there are no alternatives) and the judgement is stated. In that sense, it is circular; no reason is stated and this statement is a rephrase of the claim.

hard-case: doc #cID5288

Is a one-state solution an acceptable solution to the Israeli-Palestinian conflict?

S1 A single democratic state with equal rights for all its citizens is the only acceptable and only practical solution to the Israeli-Palestinian conflict.

S2 With one democratic state, the Palestinian refugees will be able to return to their homes, and Jews will be able to live anywhere in Eretz Yisrael (the Land of Israel).

S3 Palestinians will no longer be forced to live behind walls and be harassed at checkpoints, nor have their olive trees uprooted or homes demolished.

S4 Israeli Jews will no longer be ostracized by the world for persistent human rights violations, and political leaders will no longer be able to avoid addressing the everyday concerns of citizens by shifting the focus to the conflict.

S5 Jewish and Arab schoolchildren will receive the same level of education and have their schools funded equally by the government, and the economy will benefit from a larger pool of workers and innovators.

S6 A single democratic and secular state is the ONLY acceptable and practical solution to the Israeli-Palestinian conflict and the only way forward.

S7 By politically unifying the territory and making "nationality" and "citizenship" synonymous, Israel and Palestine will be able to truly shine to the world as a beacon of freedom and comradeship.

This is one of the texts which require analysis. The writer presents the premises as the outcomes of her claim. However, those premises or outcomes are the products wished whatever the solution is. Thus, they might also be possible with the reverse solution / non-one state solution and the writer does not show that it is not possible the other way around (non-one state situation) so she has the burden of proof. The argumentation is circular.

3.8 Conclusion

After our initial annotation study, we concluded that it is feasible to build a reliable dataset labelled for fallacies, given the increase in agreement values between succeeding annotation tasks. In this study, firstly, the experts are trained more; secondly, we elaborated on the typology of fallacies towards a finer-grained ontology and could filter out some types which are not easy to recognise or not well-defined and finally, we decided on which source to make use of to obtain the dataset considering the density of fallacious content and the level of discussions.

4 Secondary Annotation Study

In this study, we investigated the possibility of building a ground truth, a set of texts with gold-labels and the feasibility of collecting a lot of data labelled by non-experts, i.e. via crowdsourcing the annotations.

4.1 Building the Ground Truth

As given in Section 3.7, we have a set of 56 texts from the procon portal, annotated for 16 types of fallacies. In addition to that, the same annotators labelled another set of texts, 103 documents in total, from the same source; this time the number of fallacy types was 15. The guideline for the annotators can be found in Section 2 for the respective fallacy types. As in the preliminary tasks, annotation is conducted at the sentence-level. The platform of annotations is webanno. In Table 11, we present the set of labels we introduced in the annotations.

ad hominem appeal to force begging the question false cause	appeal to consequences appeal to ignorance biased sample red herring	appeal to emotion appeal to popularity composition / division ^a straw man	appeal to false authority appeal to sarcasm / ridiculising false analogy sunk costs
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^a This fallacy type is removed from the annotation task with 159 docs.

Table 11: The fallacy types introduced to the annotators for both sets.

Observations:

- Additionally given as a choice was the label *no-fallacy*.
- Out of the introduced 16 types for annotating 59 docs, 5 types are never used by any of the annotators. These are, namely, appeal to force, appeal to ignorance, appeal to popularity, false cause, straw man, sunk costs.
- Out of the introduced 15 types for annotating 103 docs, 2 types are never used by any of the annotators. These are, namely, appeal to ignorance and sunk costs.

Below in Table 12, we present the agreement values for both annotation sets.

Set	# of fallacious sentences / total # of sentences	# of categories used / total # of categories	Krippendorff's alpha	Cohen's kappa
59 docs	46 / 266	11 / 16	0.0842	0.0899
103 docs	47 / 539	13 / 15	0.151	0.1528

Table 12: Agreement results of the two disjoint gold-data sets based on Krippendorff's alpha and Cohen's kappa.

The agreement values with both measures is low; apparently the annotations are not reliable. Indeed, the agreement values at binary categorization (fallacious / non-fallacious) are similar (around 0.15 with both measures for both datasets). We can state the possible reasons as follows:

- The number of labels is too high to achieve a plausible agreement value.
- The number of sentences labelled fallacious is far less than the number of non-fallacious ones.

4.1.1 Curation

For each two datasets, we conducted a curation study. After verbal discussions about the annotations item by item, we assigned the final label for each item. As a result, we have a set of 159 texts, 805 sentences in total, with gold labels. The number of occurrences of each fallacy types in the gold labels is given in Table 13.

fallacy type	# of occurrences in annotations
ad hominem	5
appeal to consequences	4
appeal to emotion	2
appeal to false authority	4
appeal to popularity	2
appeal to sarcasm / ridiculising	3
begging the question	2
biased sample	3
false analogy	3
no-fallacy	774
red herring	2
straw man	1

Table 13: Agreement results of the two disjoint gold-data sets based on Krippendorff’s alpha and Cohen’s kappa.

4.1.2 Learning experiments

Having a fair amount of labelled instances, we run some classification experiments. We tried in both sentence-level and document-level classification settings, with binary labels (fallacious / non-fallacious). The features we used are for just trial purposes only, to name they are, character n-grams, counts of named entities, ratios of POS tags, ratio of questions and superlatives, applicable in both settings. The experiments are conducted using a tool designed for text classification tasks called dkpro-tc [Daxenberger et al., 2014], which wraps the learning algorithms implemented inside the WEKA tool [Hall et al., 2009]. We saw that the test results are unexpectedly high. The weighted accuracy and f-measure results for both settings are above 95% averaged over 10-fold cross-validation. This can be attributed to our diverse set of features and the imbalance of the labels in the dataset. No further experiments are held to see the real quality of the data or investigate the results of a coarse-grained labelling scheme. See A.1.1 for the details of the code.

4.2 Investigating the Feasibility of Crowdsourcing

Since the annotation by experts takes a long time and the reliability is not guaranteed, a faster solution is devised. The wisdom of crowds is said to reasonably contribute to the formation of a reliable annotations. Indeed, the decision by non-experts is good indicator of the preciseness of the statement of the problem at hand.

The platform this project uses is Amazon Mechanical turk ², one of the widely-used tools where some registered users annotate some task (called human intelligence task, in short HIT) that they themselves choose or are qualified to participate in. The HITs are uploaded usually for research purposes by researchers or institutions. On this platform, the annotators of HITs are called workers and they are generally paid very low ³. They are registered to amazon’s system and can take on a HIT that they choose. If the HIT requires a qualification, the worker is examined with a short set of questions. If she passes and is willing to continue, she answers the HIT and is paid according to the decision of the provider. An example HIT page in our work can be seen in Figure 3. The guidelines and task description is always on the HIT page. The fallacy type definitions are summarised from the descriptions given in Section 2 and are supported with different and shorter exemplary sentences.

² <https://www.mturk.com/mturk/>

³ See [Fort et al., 2014] for an mturk-critic discussion.

Task description

Mark the category of the "bad argument" for each sentence.

Some arguments are better than others, some make a good point and are likely to convince others, some are not that persuasive. It's usually a matter of discussion and debate.

There are however arguments which are objectively pretty bad — these usually use personal attacks, involve appealing to emotions instead of focusing on reasons, manipulate the argument and draw attention from the problem, and so on — let's call them *fallacies*.

In this task, you are given three arguments (each consisting of several sentences). **For each sentence, you have to decide whether or not people would spot a particular fallacy.** We refer to 'people' as someone with common human understanding of language as well as good common background knowledge; not necessarily an expert on the topic, but also not someone who gets fooled that easily.

Label categories

"Ad hominem" ... attacks the opponent or any other parties involved in the argument personally, instead of debating

Argument topic: Can Alternative Energy Effectively Replace Fossil Fuels

Author's stance: **pro**

Argument text (one sentence per row)	Sentence labels					
Alternative fuels most certainly can be worked into the vast majority of power sources, its all a matter of the right person taking the initiative.	☐ Ad Hominem	☐ Irrelevant authority	☐ Appeal to emotion	☐ Biased sample	☐ None of these	
I'm against nuclear power for obvious reasons, and there are plenty of other, safer alternatives.	☐ Ad Hominem	☐ Irrelevant authority	☐ Appeal to emotion	☐ Biased sample	☐ None of these	
Wind, tide and sun are good examples, tide power could be taken along flood barriers and floating structures, but I would be concerned about the creatures beneath.	☐ Ad Hominem	☐ Irrelevant authority	☐ Appeal to emotion	☐ Biased sample	☐ None of these	
Fill deserts with sun and wind farms.	☐ Ad Hominem	☐ Irrelevant authority	☐ Appeal to emotion	☐ Biased sample	☐ None of these	
Cannabis could fully replace oil and timber, and they absorb a MASSIVE amount of CO2	☐ Ad Hominem	☐ Irrelevant authority	☐ Appeal to emotion	☐ Biased sample	☐ None of these	

Figure 3: An example HIT page in our annotation task.

Before fully devising a crowdsourcing mechanism, we firstly had our gold-dataset annotated by non-expert annotators to see the quality of the annotations. Having seen from our initial annotation studies that the high number of categories causes confusion and increases the chance of disagreement between the annotators, we chose 8 fallacy types out of our set of categories. These are, namely, ad hominem, irrelevant authority, appeal to emotion, biased sample, appeal to consequences, appeal to popularity, appeal to sarcasm, straw man and additionally a category 'no-fallacy'. Since this set is still crowded, we employed two tasks for each sentence; dividing our category set into two. Thus, each sentence is annotated twice; once for the labels, which we call category1, ad hominem, irrelevant authority, appeal to emotion, biased sample; and another once for the labels, which we call category2, appeal to consequences, appeal to popularity, appeal to sarcasm, straw man. For each annotation task (per sentence), we required 3 annotators.

Overall, the it took about 10 days to have the full set of annotations. There were a total of 159 texts; the annotation items were sentences in these docs, in total 805 sentences; 106 hits created, each having differing numbers of annotation items and in total 25 annotators involved in the task.

4.2.1 Agreement among mturkers per hit

The hits have a certain set of documents whose sentences are annotated by exactly 3 annotators for either the labels of category1 or category2. Thus, we can calculate the agreement of mturkers per hit ⁴. Below in Table 14, we present the hits with the first ten and last ten highest agreement results.

⁴ Find the details of the code to parse the answers per hit and to calculate the agreement values per hit in Appendix A.1.3

hitid	nItems	krippendorff's alpha agreement
3Q9SPIIRWJYGOFKR7Z6PEHA49APAWR	13	0.8323
3I6NF2WGIG8NLNIAXYBQK38UN8Y5GH	16	0.5493
3O4VWC1GEWIUYAL7RL802J2Q14P3JU	21	0.5305
308KJXFUJRIOWYGXCU1ERX355G7AT1	18	0.5053
3ZFRE2BDQ9QWENXPTWIAXTFKRX2XZ8	7	0.5
3NFWQRSHVEQFNKBZ3AOZ6FZ7C2CFGA	19	0.4865
3DIIW4IV8P7JLVK59HSJ0NFW2PVI4H	16	0.4622
3OREP8RUT2NPA3ZJY823NP8MX4LBGP	7	0.4615
3H5TOKO3D9V3SBP47GJU3KPRTY164C	24	0.4577
3UYRNV2KITBCIXIYI0VOM7RIQALN89	18	0.455
..	..	..
39AYGO6AFFWPEJZAPLYQ2TVRK FVN6N	24	-0.1899
3538U0YQ1F6ET8ZB8VHFNLAY8X83FR	13	-0.2053
3J94SKDEKI1K938JV8AIEI5XAGI5D7	33	-0.2264
324N5FAHSXN9L3T8TGEIMTUM7DHKV6	25	-0.2337
3W9XHF7WGK7U8BPJOLE3LB61AS1KTQ	6	-0.2347
3RKHNXPBGW898O48OQ4DQ7S3T68KUH	15	-0.2446
3LVTFB9DE5UCE918HQZTIE3LJ4MGQ6	10	-0.25
3VO4XFFP15YTLGQ7ONVZC1QSK12Q7C	45	-0.2535
3G5RUKN2ECFCW218NS12M3MTAG3N9Y	18	-0.2541
3SV8KD29L44QUNYIUD3L7Z7IVONKZ6	22	-0.2914

Table 14: The highest 10 and the lowest 10 agreements results out of 106 hits, based on Krippendorff's alpha measure.

4.2.2 Detecting spammer annotators based only on the gold labels

We applied a simple quality measurement mechanism to evaluate the annotations of mturkers. As explained previously, we have the gold labels for each annotation item (sentence). And by the crowdsourced annotations, for each item, we have three answers for each category (category1 or category2). Applying simple majority voting, for each item we get two labels, one label for category1 and another label for category2. On majority voting, we assign to the item the label that is given by two or all the annotators, if all the three annotors gave different answers, then the label assigned is 'unknown'. As for the expert labels which are not included in crowdsourcing task (we have a total of 8 labels in mturk annotations while we had 11 gold labels), we just assigned 'no-fallacy' to the expert label that is not included in the mturk setting. Then, for each annotator, we calculated two disagreement values, between her and majority, and between her and gold labels, respectively. This disagreement value is the number of the non-matching answers of the annotator (with respect to majority or gold label) divided by the total number of annotations given by that annotator. Below in Table 15 we present the disagreement rates per mturker.

We conjectured that those with high disagreement values are more likely to be spammers. Skimming through the answers of the annotators, we have found that four annotators (see the last four rows in Table 15) are indeed giving random or constant answers in a visible pattern. Thus, we rejected the annotations of these four annotators.

In addition to the disagreement values per annotator, we calculated the agreement values between majority and gold labels using both Krippendorff's alpha and Cohen's kappa measures. Treating each category as a distinct annotation task, given we have gold labels and majority mturker decisions for each annotation item, we can get the chance-corrected agreement results, which are also informative. Indeed, we also calculated agreement with and without spammer answers to examine their effect to the reliability. Below in Table 16 we present the values.

	nItems		Krippendorff's alpha		Cohen's kappa	
	category1	category2	category1	category2	category1	category2
full set of annotations	805	805	0.0948	0.0962	0.1189	0.1117
no spam annotations	169	154	0.0551	0.0688	0.1122	0.1004
only spam annotations	636	651	0.1063	0.1007	0.1223	0.1131

Table 16: The agreement values between majority mturker decision and gold labels for each category.

According to the agreement values in Table 16:

→ The spammers' agreement is higher than that of non-spammers and that of the mixed set.

annotatorid	nAnnotations	weightedMajorityDisagr	weightedGoldDisagr
A2TBXASXZIRNNW	998	0.1253	0.0832
A2ZUKYMM3YV6JN	1290	0.138	0.1163
A2SD9BHIIIVGLX	41	0.2195	0.1463
A1GYADT8LQP9D5	47	0.1064	0.1489
AO2V9VVYAL7OK	368	0.231	0.1875
ABM77ZQWCHPNX	77	0.2597	0.2208
A3VIBIROKRS6Z9	1223	0.2093	0.224
A3TQB2CTVLTGN5	80	0.275	0.2375
AODCAY9I2MBXV	25	0.24	0.24
A1FBBY2JJYRMRI	12	0.3333	0.25
A3GNQDFPZALU92	6	0.3333	0.3333
A99PQSI7FTY50	9	0.2222	0.4444
A2TQZQYR1X4G56	13	0.3846	0.4615
A26XN1TM6CV01T	57	0.4561	0.4912
A36LOA6VLJU157	12	0.1667	0.5
A1WA3VMWGGJPYS	6	0.5	0.5
A1BHDJMSL5EKDB	16	0.5625	0.5
A1JMB9Z7EYHU1O	250	0.596	0.572
A2NNCGHJVK0L1K	16	0.625	0.625
A17EHWQND2HPWE	122	0.5902	0.6393
A31BIDN8JJS14R	11	0.7273	0.7273
A3NKM6RL2KVUB6	20	0.75	0.75
A1DE0J98KFXN0Z	14	0.7857	0.8571
AQ9F0W1HKUF2J	282	0.8617	0.8582
A34CWKC6HUSM5B	21	0.7143	0.9524

Table 15: The disagreement rates of mturkers between the majority vote and between the gold labels, respectively.

- The spammers who participated in the hits of category2 are more disturbing, while the spammers involved in the hits of category1 can be said to be more consistent or professional.
- Krippendorff’s alpha measure is more sensitive to the change in the set of annotations, in our setting.
- As a whole, our expectation that after removing the spam annotations, the agreement values will increase is refuted.

4.2.3 Estimating the Quality of Annotations

Since the agreement values are reversely effected by spammer involvement contrary to our expectation, the values are not informative in terms of the quality of annotations. Thus, we need a better measure or a mechanism to estimate. In Section 4.2.4 we present the short survey we collected from the literature on annotation crowdsourcing studies. Out of the models we examined, we have tried the probabilistic model of [Ipeirotis et al., 2010] to get an estimate of the quality of our mturk annotations. However, the results were not quite interpretable, mostly negative numbers and generally various similar values from many different measuring mechanisms. Finally, we applied the model of [Hovy et al., 2013]. It outputs a ranking for annotators and the predicted labels for the annotation items. In Tables 17 and 18, respectively for our two categories, we present the results related to the annotator quality estimates. As can be seen, the ranking is different than that of disagreement values based on expert labels; still, we can say that spammers’ quality values are low.

annotatorid	quality estimate
A3TQB2CTVLTGN5	0
A1GYADT8LQP9D5	0.0002
A2SD9BHIIIVGLX	0.0003
A2TQZQYR1X4G56	0.0003
A1DE0J98KFXN0Z	0.0005
A3NKM6RL2KVUB6	0.0005
A34CWKC6HUSM5B	0.0006
AQ9F0W1HKUF2J	0.0142
A2TBXASXZIRNNW	0.0397
AO2V9VVYAL7OK	0.045
A1JMB9Z7EYHU1O	0.0462
A2ZUKYMM3YV6JN	0.0557
ABM77ZQWCHPNX	0.0795
A1FBBY2JJYRMRI	0.0855
A3VIBIROKRS6Z9	0.1916
A1BHDJMSL5EKDB	0.2367
A17EHWQND2HPWE	0.2472
A3GNQDFPZALU92	0.2538
A1WA3VMWGQJPYS	0.318
A36LOA6VLJU157	0.5687

Table 17: Quality rank of annotators in category1 according to [Hovy et al., 2013].

annotatorid	quality estimate
A26XN1TM6CV01T	0.0001
A2SD9BHIIIVGLX	0.0001
AODCAY9I2MBXV	0.0002
AQ9F0W1HKUF2J	0.0003
A31BIDN8JJS14R	0.0003
A3NKM6RL2KVUB6	0.0004
A2NNCGHJVK0L1K	0.0004
A34CWKC6HUSM5B	0.0006
A2TBXASXZIRNNW	0.0327
ABM77ZQWCHPNX	0.0396
A1JMB9Z7EYHU1O	0.0549
A2ZUKYMM3YV6JN	0.0578
A1GYADT8LQP9D5	0.0658
A3VIBIROKRS6Z9	0.1283
A17EHWQND2HPWE	0.1313
A3TQB2CTVLTGN5	0.1682
AO2V9VVYAL7OK	0.2267
A99PQSI7FTY50	0.3796

Table 18: Quality rank of annotators in category2 according to [Hovy et al., 2013].

4.2.4 A Survey on Measuring the Quality of Crowdsourced Annotations

Below we give a short listing for the literature we surveyed related to the annotation quality measures.

- [Filatova, 2012]:
 - Doc-level annotation for irony and sarcasm on amazon product reviews using mechanical turk.
 - Claiming that there is no strict definition of irony, they apply crowdsourcing and on purpose do not provide any guidelines to the annotators.
-

-
- Having a set of paired reviews for each product (one review sarcastic and the other non-sarcastic) by the mturkers, they have this collection re-annotated by different mturkers; each review annotated by 5 annotators.
 - To filter out spam annotations, they apply 1) majority voting 2) Quality management algorithm designed by [Ipeirotis et al., 2010], specifically for mturk annotations
 - The paper points to good resources for quality management.
 - [Aroyo and Welty, 2013]:
 - They consider disagreement is useful and informative: "Annotator disagreement is not noise but signal". They use disagreement information for rating sentence clarity / ambiguity as well as annotation quality.
 - The task on the stage is relation extraction from sentences. The annotation is sentence based and crowd-sourced.
 - Related to our search of quality measure, they apply classification for identifying low quality annotators. The features they use are 1) worker-sentence disagreement 2) worker-worker disagreement and 3) average relations per sentence. Their linear classifier, for which we don't find much detail in the paper, performs with an accuracy of 98% so it outputs 12 low-quality annotators out of 110.
 - The authors later develop a framework for assessing quality, Crowd Truth detailed in the paper [Inel et al., 2014].
 - [Sarasua et al., 2012]
 - They propose a framework capable of combining different crowdsourcing platforms
 - Their analysis of crowd annotations is rather qualitative.
 - [Nowak and Rüger, 2010]
 - The paper is a bit less-organised in terms of wording and research questions but findings and methods are very informative.
 - They work on image data which is annotated by both experts and mturkers for concept labels.
 - They measure two following: 1) Agreement among experts using accuracy and majority vote 2) Agreement among mturkers 3) Agreement between expert and non-expert (mturker) annotations 4) Influence of annotations on system ranking
 - They measure the quality of annotations using some metrics specific to image analysis studies. Thus, the annotations are ranked for their quality. Then, the correlation between the ranked lists are evaluated using Kendall's tau statistic for obtaining "the degree of correspondence between the two rankings" and then using Kolmogorov-Smirnov's D for assessing "the significance of this correspondence", each referenced in the paper. This part was interesting in that if it is persuasive in terms of evaluation, then we may apply them as well and study the details more.
 - Their most apparent finding was that after applying the measures, showing that majority vote is able to filter out noisy annotations in the mturk annotations. So they say it is safe to use majority vote after showing that the quality of mturk annotations are comparable to that of expert annotations.
 - [Hirth et al., 2011]
 - For cheat detection, the authors propose two mechanisms: 1) agreement with the majority decision 2) checking with a control group.
 - The 1st mechanism is interesting. The authors construct a model with priors from the probability that the annotator is a cheater and the posteriors from the probability that the annotator's answer is wrong. They conjecture that the number of incorrect results follow a binomial decision. If the incorrect results is 50% of all answers, the majority decision is correct.
 - For the 2nd mechanism, they have their mturk-annotated items validated by supposedly another set of mturkers.
 - They claim the two mechanisms infer the same information about quality; in their own words "both majority decision and control group offer the same quality of cheat detection quality".

- [Heilbron and Niebles, 2014]

-
- They perform an activity annotation task. The aim is to show that mturk is suitable for data collection for such a task.
 - They have both single-expert and mturk annotations. Measuring precision, recall and f-score of mturk annotations against ground truth, they find the values are generally above 95% and state that the task and the proposed setting is suitable for collecting data with the help of crowdsourcing.
 - [Laws et al., 2011]
 - They apply active learning to the instances to be annotated on a crowdsourcing platform to increase reliability of the annotations and the efficiency of the process.
 - They claim and show with their experiments on different working domains that active learning and crowdsourcing are complementary tasks in a ground-truth formation process.
 - [Ipeirotis et al., 2010]
 - They propose a probabilistic model for estimating 1) worker quality 2) item labels. The model is built upon an EM-based algorithm by [Dawid and Skene, 1979] with a contribution that takes annotator bias into account in addition to the possibly systematic wrong answers (spamming).
 - They show that their algorithm increase the quality of annotations (eliminating low-quality annotators after detecting them with this algorithm) by 3% and decreasing the cost by 30% in an experiment that they conducted with real data.
 - [Hovy et al., 2013]
 - They develop a generative model to estimate 1) worker quality 2) item labels.
 - The model is fully unsupervised. The true labels of the items are assumed latent variables, uniformly distributed initially. Another assumption is for the spamming behaviour of the annotators. Being spammer is taken a binary variable and is picked from a Bernoulli distribution with a parameter, denoting the trustworthiness of the worker and is to be learned. If the annotator is not spammer, then she chooses the true label. Otherwise, she behaves like a spammer, where the behaviour is modelled as a vector drawn from a multinomial distribution whose parameter is also another one to be learned.
 - Spamming behaviour is assumed to be independent from the true label, therefore, the authors say the bias dimension is not taken into account and the trustworthy annotators can be discriminated from the malicious spammer ones, while the model is kept simple in the meantime.
 - They find that the model captures spamming behaviour well such that when spammer answers detected by the algorithm removed, the predicted labels' correlation with true labels is considerably high.
 - They also show that their model outperforms majority voting and is comparable to some other probabilistic quality models, in terms of accuracy, evaluated for the label prediction capability of the model.

4.3 Conclusion

During the initial crowdsourcing study, we observed that firstly annotation of fallacies by non-experts is not impossible or ineffective; on the contrary, the annotators can very well capture the flaws, at least in our domain of user-generated content streams. Furthermore, the some annotators indicated how they found the task interesting via their comments on the site. Secondly, although the crowdsourcing is not impossible, the agreement values are low, which can still be attributed to the difficulty of the task. At the same time, this also shows the ill-defined state of the task we proposed. And another factor was that the instances to be annotated are highly imbalanced; the dataset contains far less non-fallacious instances than the data contains few fallacious ones which is said to affect the agreement values. Therefore, we have to look for ways to improve the fallacy rate of the data and propose to the annotators a dataset which is likely to contain mostly, and preferably only, fallacious instances.

5 Crowdsourcing the Annotations

In this part, our objective is to have a large set of comments annotated on a crowdsourcing platform by non-expert people. The texts will be annotated for five types of fallacies, namely, 1) ad hominem 2) appeal to emotion 3) appeal to sarcasm 4) appeal to authority 4) appeal to consequences 5) hasty generalization (this comprises biased sample and appeal to popularity). The comments are user thoughts submitted to some online discussion portals on some controversial topics; these texts are usually context and dialogue-dependent and are informal in tone. From our previous studies, we have found using crowdsourcing for having the texts is feasible and we will use this medium for the annotations again. Before that, we have to increase the chance of having as many fallacious instances as possible.

5.1 Obtaining the Texts Likely-To-Be-More-Fallacious

Previously we had a set of 159 texts annotated twice; once by two experts, once by mechanical turk annotators. The quality of these annotations was not high; the agreement was low in both cases. This can be attributed to the difficulty of the task, poor definition of the task and to the suitability of the data, basically. As regards to the suitability of the data, we saw that the label distribution was reasonably imbalanced; a prevailing rate of the texts / sentences was already not fallacious, which effects the chance for a correct labelling of a sentence. For this reason, we run a simple heuristic to have in the annotatable set of texts as much fallacious data as possible.

Distant supervision is a method, like bootstrapping, for enlarging the set of labelled data using some heuristic with little automated or manual intervention, first proposed by [Mintz et al., 2009] for the task of relation extraction. In this study, we apply a fairly simple distant supervision method to filter out fallacious comments from our large raw dataset.

Our data instances are textual comments written by some registered users for a specific topic, in reply to some other user or an article, on some online portal. Each file retrieved from a portal is for a discussion thread, containing several comments in it, having internal parental relationships (response). We have two online sources:

- debate portals: The portals createdebate.com and convinceme.net were crawled by IH previously. From these, we have 24325 discussions retrieved, having a total of 411521 comments. The dump files are in xml format⁵.
- Disqus (disqus.com): The portal that gathers many discussion pages from a number of sites was used as a source in [Bamman and Smith, 2015]; we crawled this portal using a modified version of the crawler kindly provided by these authors. We could retrieve 142654 discussion threads, having roughly more than 3M comments. The dump files are in json format⁶.

5.1.1 Selecting Fallacious Comment Candidates

The aim is to collect the comments which are candidate to be fallacious. We conjecture that if a comment X in some discussion thread t is accused of being fallacious by its child comment A (the response), then comment X is a candidate to be fallacious. By ‘accused of being fallacious’, we mean that the child mentions a fallacy name so it is our intuition or reduction that this mention is towards the parent comment like saying ‘you commit the fallacy of *ad hominem*’, for example. Thus, we need a set of seed keywords for the types of fallacies we are going to annotate. We call the process ‘candidate selection’. Below we present the algorithm in 1.

For the code details, see A.1.2.

In Table 19, we present the set of keywords for different fallacy types. Those rows with a star (the first 6) make the set of keywords that we actually use in our crowdsourced annotations. Nonetheless, we looked for other keywords to see how our distant supervision procedure performs and how our selected fallacy keywords are distributed. We don’t apply grouping but take the keywords as a list combining all the items in the rightmost column, having a total of 47 keywords. The grouping below is for clarification purposes but of course technically can be applied easily.

⁵ Available on [smb://fry.ukp.informatik.tu-darmstadt.de/data_repository/Fallacies/disqus/](http://fry.ukp.informatik.tu-darmstadt.de/data_repository/Fallacies/disqus/)

⁶ Available on [smb://fry.ukp.informatik.tu-darmstadt.de/data_repository/Fallacies/debateportals/](http://fry.ukp.informatik.tu-darmstadt.de/data_repository/Fallacies/debateportals/)

Algorithm 1 Distant supervision algorithm for selecting candidate fallacious comments

1: Collect comments per fallacy type where the comments are supposed to be committing that fallacy type if its child comments contains a keyword for the fallacy.

Require: `commentList`: list of comments retrieved from an online source; `keywords`: list of fallacy keywords.

```
2: for each keyword in keywords do
3:   keyword.candidateCommentList  $\leftarrow$  [] ▷ initialize a list per keyword
4: end for
5: for each keyword in keywords do
6:   for each comment in commentList do
7:     if keyword.string occurs in comment.text then
8:       parentComment  $\leftarrow$  comment.parent
9:       if parentComment is not null then
10:        if parentComment.parent is null then ▷ discard the candidates which are deep in the discussions
11:          keyword.candidateCommentList.append(parentComment)
12:        end if
13:      end if
14:    else
15:      keyword.singleReferrentCommentList.append(comment) ▷ keep this list for further use, maybe for
16:    end if
17:  end for
18: end for
```

fallacy name	keywords
fallacy*	fallacy, fallacies, fallacious
ad hominem*	ad hominem, ad hominam, tu quoque, ad feminam, ad feminem, guilt by association
appeal to emotion*	appeal to emotion, appeal to pity, ad odium, appeal to spite, ad passiones
appeal to sarcasm*	appeal to sarcasm
appeal to irrelevant authority*	appeal to authority, appeal to irrelevant authority, ad verecundiam, argument from authority
appeal to consequences*	ad consequentiam, appeal to consequences, slippery slope, sunk costs
hasty generalization*	hasty generalisation, hasty generalization, faulty generalization, biased sample, appeal to popularity, ad populum
false analogy	false analogy
false cause	false cause
straw man	straw man
begging the question	begging the question, circular reasoning
appeal to force	appeal to force, ad baculum
red herring	red herring, redherring
appeal to ignorance	appeal to ignorance, ad ignorantiam
trivializing	trivializing, trivialising
bandwagon fallacy	bandwagon fallacy
non sequitur	non sequitur
two wrongs don't make a right	two wrongs don't make a right, two wrongs do not make a right, two wrongs donot make a right

Table 19: The keywords per fallacy type input to the distant supervision algorithm.

On both raw collections, we run our algorithm with these 47 keywords. The results can be found in Table 20.

keyword	sources	
	disqus	debate-portals
ad baculum	2	0
ad hominem	1186	39
ad ignorantiam	3	1
ad odium	1	0
ad populum	19	14
ad verecundiam	5	0
appeal to authority	58	10
appeal to consequences		2
appeal to emotion	44	10
appeal to force	1	0
appeal to ignorance	7	1
appeal to pity	3	0
appeal to popularity	5	2
argument from authority 13		3
bandwagon fallacy	4	1
begging the question	63	27
circular reasoning	32	20
fallacies	441	46
fallacious	368	71
fallacy	1751	215
false analogy	59	2
false cause	14	0
faulty generalization	4	0
guilt by association	150	3
hasty generalisation	0	4
hasty generalization	25	12
non sequitur	409	15
red herring	2	29
slippery slope	909	65
straw man	1300	24
sunk costs	9	0
trivialising	2	0
trivializing	44	1
tu quoque	107	2
two wrongs do not make a right	24	4
two wrongs don't make a right	128	9
<i>total # of candidate comments</i>	<i>8052</i>	<i>632</i>
<i>total # of comments</i>	<i>27813683</i>	<i>411521</i>
rate	0.00029	0.0015

Table 20: The number of comments extracted as candidate to be fallacious from each source.

→ To see the quality of the texts extracted this way, we annotated a set of 20 comments for each keyword (keywords of those 5 types we are going to finally use on mturk). There is generally a problem of missing context as the comments are written for a piece of larger opinion text or news article. However, it is not hard to get the general idea and flow so it is feasible to have these texts annotated by non-expert people.

5.2 Results from the Crowdsourced Annotations

5.2.1 Annotation results on commentset

The dataset is previously collected and used by [Habernal and Gurevych, 2012]. The texts are comments by users from a social forum on the web. Each text is split to sentences and deployed to mturk to be annotated for fallacies. As opposed our previous mturk task, here the set of fallacies is diminished to have only 5 types, which are, ad hominem, appeal to consequences, appeal to emotion, appeal to sarcasm and hasty generalization (this comprises biased sample and appeal

to popularity) and additionally a choice as ‘no-fallacy’. The aim of this particular annotation task is to filter out non fallacious text and have the fallacious ones re-annotated in a later task which is expected to give highly-agreed more reliable annotations.

The process took around 3 weeks (314 hours in total); average time to complete a task for an mturker was 84 seconds. 2398 sentences out of 289 documents submitted; 289 hits created and 31 mturkers took part in the task, hits having and the annotators annotating varying numbers of items.

As we did previously, we calculated agreement of mturkers per hit. The results can be seen in Table 21.

Agreement among mturkers per hit

hitid	nItems	ncoders	krippendorff's alpha agreement
3XT3KXP24ZAX0IOJS5LYXLZA11O6IM	10	3	-0.45
3CVBEMEMMXBSG8NR45X853NYRWZU7HU	7	2	-0.4444
37M4O367VJUNDXEW819LX8SRMFXM52	5	3	-0.4
3FULMHZ7OU95HEAWUWQBSGMDMJ34MG	7	3	-0.3333
3OEMX9PEVKVTTBC4ZL9DB6AFBUBKSY	2	3	-0.25
..	..	..	..
3WGCNLZJKFKLL70UL86BSSG1L1D1U	6	3	0.4688
3P520RYKCHI6Q5J5H307OI9LZ10U52	9	3	0.48
3G9UA71JVV6CZTMUTXBQF0E69RY7JZ	11	3	0.4839
3SV8KD29L44QUNYIUD3L7Z7IY9AZKH	13	2	0.569
3DWNFENNE37FG64BRZDJCOUWIP84JC	3	3	0.6

Table 21: The highest 5 and the lowest 5 agreements results out of 289 hits, based on Krippendorff's alpha measure.

In Table 21, we see that the agreement values are not as high as those of our initial validity mturk task (See Table 14 in Section 4.2).

Estimates on annotator quality and item labels

In this part, what is investigated is the predicted labels of the sentences. We run the estimate model called MACE by [Hovy et al., 2013]. Those sentences and the documents they are contained in, which are predicted by MACE to be fallacious are set aside to be annotated later in an mturk task. According to the MACE model, in this annotation task, 929 sentences were predicted to be fallacious so the rate of (predicted-to-be) fallacious sentences is 0.39. Thus, 211 documents with a total of 1906 sentences can be annotated again such that each of these documents contain at least one fallacious sentence. The details of the code can be found in Appendix A.1.3.

6 Evaluation

The project plan proposed at the outset of my internship was not followed except the initial annotation studies. The project as a whole was not a research project for me but rather a programming work at some company. It was a disappointment from my side but still I tried to enjoy its scientific juice as much as possible.

A Appendices

A.1 Code

A.1.1 Learning experiments on Gold Data

The related package is `xxx.argumentation.fallacies.classification`. `FallaciousDocClassifier` and `FallaciousSentenceClassifier` are the shell classes. Both require two parameters: 1) path to the folder that contains the documents in xmi format and are labelled for fallacies in our Fallacy typesystem (See the package `de.tudarmstadt.ukp.dkpro.argumentation.types`).

A.1.2 Distant supervision

Reading the dump files and selecting the candidate comments

The files are located in the package `xxx.argumentation.fallacies.distantSupervisionV3`. The classes are:

- **Comment**: models a piece of comment
- **CommentList** (extends ArrayList): list of **Comment** objects.
- **ExtractCandidateComments**: Implements the distant supervision algorithm.
- **Dump2CandidateList** (abstract): Models a dump file that has many comments in it. Locates the comments in the relevant object and calls the **ExtractCandidateComments** to implement the distant supervision algorithm. The output candidate comments are written to the relevant locations on the disc.
- **JsonDump2CandidateList**, **XmlDump2CandidateList** (extends **Dump2CandidateList**): Overrides the reader function for specific dump file formats (json and xml, respectively, in this case). The reader is dependent on the dump file format in terms of tags.

How to run:

- for json dump files from disqus: run **JsonDump2CandidateList**
- for xml dump files from debate-portals: run **XmlDump2CandidateList**
- params:
 - param #0: Path of the file that contains a keyword in each line
 - param #1: Path to the folder where JSON files are stored
 - param #2: Path to the folder where the output file will be written to.
 - param #3: Name of this task which will be the name of the output folder under param #2.
 - param #4: Name of the website/portal the dump is taken from.
 - param #5: Format of output keywords files (write xml or txt).
 - param #6 * (applicable only to **JsonDump2CandidateList**) : Path to the folder where the thread meta-data containing topic information of the discussions is stored so that we keep the topic information of the comment as a metadata.
- Under the relevant folders, the candidate comments per keyword can be found as an xml file.

Converting the xml files containing the selected candidates per keyword to xmi files per comment

The files are located in the package `xxx.argumentation.fallacies.mturk.xml2xmi`. The classes are:

- **XmlToJcasPerComment**: Creates a new jcas for the comment item read from the input xml file, records it as an xmi file under the relevant folder.
- **PipelineXmlToXmi**: Runs the dkpro pipeline [Eckart de Castilho and Gurevych, 2014] that calls **XmlToJcasPerComment** to complete the process for multiple xml files.

How to run:

- Run **PipelineXmlToXmi**.
- params:
 - param #0 : Path to the folder that contains xml files for each fallacy type.
 - param #1 : Path to the folder where xmi per fallacy type will be written to.
 - param #2 : Path to the folder where xmi per comment will be written to.
- Under the relevant folders, the xmi file per candidate comment can be found. This can be further input to the pipeline for having them placed as mturk hits on amazon crowdsourcing web application.

A.1.3 Analysis of crowdsourced annotations

The files are located in the package `xxx.argumentation.fallacies.mturk.agreement.xmi`. The classes are:

- **MturkResultsAnnotator**: Requires 1) the csv file containing mturk annotations, exported from amazon mturk server 2) the xmi files per document which are uploaded to mturk to be annotated. Per annotation item (sentence in our case), finds the label by each annotator in the csv file and records these labels per annotation item in the xmi files of the document.
- **MturkCsvUtils**: Utility functions for processing the csv file containing the mturk annotations.
- **AnnotatedXmiAnalyser**: Requires a collection of xmi files with mturk annotations per document. Finds 1) agreement per mturk hit 2) lists annotator quality ranking and predicted labels per annotation item using the MACE algorithm by [Hovy et al., 2013]. Separately records the list of documents that contain at least one sentence predicted to be fallacious by the MACE algorithm.
- **PipelineMturkAnnotationsAnalysis**: Takes a set of xmi files and a csv file with annotations, outputs the analysis of these annotations, records the results under respective folders.

How to run:

- Run `PipelineMturkAnnotationsAnalysis`.
- params:
 - param #0 : Path to the folder where the raw xmi files are stored.
 - param #1 : Path to the folder where the annotated xmi files will be stored.
 - param #2 : Path to the csv file containing mturk annotations.
 - param #3 : Path to the folder where the output will be written to.

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